

TIER 2 · US\$199 · PROFESSIONAL LICENCE

AP Agent OS — Professional

The operating system for AP agents that earn responsibility.

Evidence Room — AP Agent OS Professional

00 — Professional Operating System

Product: Evidence Room AP Agent OS — Professional
Brand idea: Proof before permission.
Line: Agents earn responsibility. Evidence decides.
Audience: Head of AP, Controller, Transformation lead, Control owner, Finance Systems
ERP stance: Agnostic. Map field names locally. Do not copy another company's workflow.
Examples: ACME Manufacturing — **ILLUSTRATIVE** only.
Version: 1.0
Classification: Navigation and START HERE path. This file does not replace charters, controls, or test packs.

Position

Professional is the operating system for a **governed** accounts payable agent programme: sixteen named agents, a human accountability model, a control and exception language, a measurement dictionary, and a test path that earns each increment of responsibility.

It is not a prompt pack. It is not a guarantee of savings, ROI, fraud detection, statutory compliance, or autonomous payment.

If you need a facilitated enterprise pack (workshop, training, steering, change), use **Team Edition** (`../04_AP_AGENT_OS_TEAM/`). If you need a paid, time-boxed blueprint service, use **Custom Blueprint** (`../05_CUSTOM_BLUEPRINT/`).

START HERE — path 01 to 12

Do these in order for the **first increment**. Skipping a step to “get to the agent” produces an unauthorised next step.

Step	What you should do	Primary files	Exit evidence
01	Read this OS. Appoint human Accountables and deputies. Name the Evidence Room store.	This file; <code>Process_Mapping/04_RACI_STANDARD.md</code> ; <code>Templates/RACI.md</code> ; <code>Governance/00_GOVERNANCE_FRAMEWORK.md</code>	Named incumbents; store path
02	Freeze the first slice as an allow-list (one entity, one channel, one invoice type, value cap). Write what is out.	<code>Agent_Library/00_AGENT_STACK_OVERVIEW.md</code> (Monday-morning start); <code>Templates/IMPLEMENTATION_PLAN.md</code>	Slice sheet

Step	What you should do	Primary files	Exit evidence
03	Discover as-done work. Do not design the agent yet.	Process_Mapping/00_METHODOLOGY.md Step 1; 01_PROCESS_DISCOVERY_GUIDE.md ; Templates/ PROCESS_DISCOVERY.md	Signed coverage
04	Transcribe, extract, structure : current-state SOP, maps, exception codes, RACI.	Methodology Steps 2-4; 03_SOP_GENERATOR_FRAMEWORK.md ; Templates/ SOP_TEMPLATE.md ; Controls/ 03_EXCEPTION_TAXONOMY.md	Versioned process pack
05	Charter one agent at L0 . Deterministic vs model split. Forbidden tools listed.	Agent_Library/19_AGENT_CHARTER_STANDARD.md ; matching AGENT_0X ; Templates/AGENT_CHARTER.md ; Governance/ 01_AUTONOMY_POLICY.md	Signed charter
06	Complete control matrix rows, risk assessment, SoD/identity intent.	Controls/00_CONTROL_FRAMEWORK.md ; 01_CONTROL_MATRIX.md ; 02_RISK_REGISTER.md ; Templates/RISK_ASSESSMENT.md	RA + matrix rows
07	Freeze baseline KPIs and the cost model (even if AI cost is zero).	KPI_and_Measurement/00_KPI_FRAMEWORK.md ; 01_KPI_DICTIONARY.md ; Business_Case/ 01_COST_PER_INVOICE.md ; 00_BUSINESS_CASE_MODEL.md if funding	Baseline ID; CPI version
08	Historical test on a labelled pack you own.	Testing/00_TESTING_STANDARD.md ; 01_HISTORICAL_TEST_SCRIPTS.md	Test Go / Hold / Redesign / Stop
09	UAT — can named people operate the path?	Testing/02_UAT_TEMPLATES.md ; Templates/ UAT.md	UAT Go
10	Shadow — live compare, no action verbs .	Testing/03_SHADOW_MODE.md	Shadow recommendation
11	Controlled pilot of named verbs only , then Measure (four families, no netting).	Testing/04_PILOT_METHODOLOGY.md ; Governance/ 02_RELEASE_AND_CHANGE.md ; 03_INCIDENT_AND_OVERRIDE.md ; KPI_and_Measurement/02_SCORECARD_GUIDE.md ; 03_WEEKLY_AGENT_REPORT.md ; 04_MANAGEMENT_DASHBOARD.md	Pilot exit + Measure pack
12	Expand one increment or Hold / Roll back / Redesign. Operate the weekly/quarterly cycle.	Methodology Step 10; Autonomy policy; Business_Case/03_IMPLEMENTATION_ROADMAP.md Phase 10; 18_HUMAN_VS_AGENT_DECISION.md	Change record

Typical first agents: 01 Intake, 10 Duplicate, 16 Orchestrator at L0 — then 02/03. **Leave 12**
Payment at recommend/prepare. Payment release, bank-change, policy exception, and legal dispute stay human from day one.

Time: a **well-bounded** first agent may fit an **ILLUSTRATIVE** 4–6 week observe-to-shadow path (Business_Case/03_IMPLEMENTATION_ROADMAP.md §4) with explicit caveats. A multi-entity stack will not.

Module map

Agent Library

File	Use
Agent_Library/00_AGENT_STACK_OVERVIEW.md	Sequence, packet, human classes, Orchestrator
AGENT_01 ... AGENT_16	Charters
17_AUTONOMY_PROGRESSION.md	Only promotion method
18_HUMAN_VS_AGENT_DECISION.md	Line-call tree
19_AGENT_CHARTER_STANDARD.md	Quality bar

Process Mapping

File	Use
00_METHODODOLOGY.md	Observe → Expand (ten steps)
01_PROCESS_DISCOVERY_GUIDE.md	As-done discovery
02_TRANSCRIPT_TO_WORKFLOW.md	Transcript and extract
03_SOP_GENERATOR_FRAMEWORK.md	Structure artefacts
04_RACI_STANDARD.md	Accountability rules

Governance

File	Use
00_GOVERNANCE_FRAMEWORK.md	Accountability, SoD, privacy, logs, vendors
01_AUTONOMY_POLICY.md	L0–L4, one increment
02_RELEASE_AND_CHANGE.md	C0–C4, hashes
03_INCIDENT_AND_OVERRIDE.md	Contain, override, fallback, BCP

Controls

File	Use
00_CONTROL_FRAMEWORK.md	Control design
01_CONTROL_MATRIX.md	Control rows
02_RISK_REGISTER.md	Risk language

File	Use
03_EXCEPTION_TAXONOMY.md	EX-* codes

KPI and Measurement

File	Use
00_KPI_FRAMEWORK.md	Four families; no netting
01_KPI_DICTIONARY.md	If it is not here, it is not a KPI
02_SCORECARD_GUIDE.md	How to score a period
03_WEEKLY_AGENT_REPORT.md	Operating cadence
04_MANAGEMENT_DASHBOARD.md	Steer view

Testing (this increment of the product)

File	Use
Testing/00_TESTING_STANDARD.md	Binding test discipline
Testing/01_HISTORICAL_TEST_SCRIPTS.md	Labelled historical pack
Testing/02_UAT_TEMPLATES.md	People operability
Testing/03_SHADOW_MODE.md	Live, no action
Testing/04_PILOT_METHODODOLOGY.md	Narrow reversible action

Business Case

File	Use
Business_Case/00_BUSINESS_CASE_MODEL.md	Inputs/outputs; Conservative / Base / Upside
Business_Case/01_COST_PER_INVOICE.md	KPI-FIN-CPI build
Business_Case/02_AGENT_ECONOMICS.md	Envelope; CCO
Business_Case/03_IMPLEMENTATION_ROADMAP.md	Phase 0-10; 4-6 week caveats

External research (Deloitte 8 Oct 2025 — 63% AI deployed / 21% measurable ROI; Hackett Nov 2025 software-adopter 60% touchless / 3.5× productivity at ≥30% touchless) may appear **labelled as research, not Evidence Room results**. They are not default targets.

Templates

File	Use
Templates/SOP_TEMPLATE.md	Example + blank SOP
Templates/AGENT_CHARTER.md	Example + blank charter
Templates/PROCESS_DISCOVERY.md	Observation working papers

File	Use
Templates/RACI.md	Operating + agent-task RACI
Templates/UAT.md	Blank UAT pack
Templates/RISK_ASSESSMENT.md	Slice risk assessment
Templates/MEETING_GUIDE.md	Decision meetings
Templates/IMPLEMENTATION_PLAN.md	Single-increment plan

Operating rule (every artefact)

Each working section answers:

1. **What** should I do?
2. **How**?
3. **Who** owns it?
4. What can go **wrong**?
5. How do I **control** it?
6. How do I **measure** it?
7. What **evidence** proves it?

What stays human

Payment release · Vendor bank-change approval · Policy exceptions · Legal disputes.

Confidence may narrow action. It may not widen a control. Agents are never Accountable.

What this OS is not

- A promise that cycle time or cost will fall.
- A compliance certificate or fraud-detection product.
- A licence to bypass ERP controls, DOA, or SoD.
- Permission to process payments without the buyer's authorisation path.

Proof before permission. Agents earn responsibility. Evidence decides.

End of 00_PROFESSIONAL_OPERATING_SYSTEM.md

AP Agent OS — Agent Stack Overview

Product: Evidence Room / AP Agent OS

Brand idea: Proof before permission.

Line: Agents earn responsibility. Evidence decides.

This library is the operating design for a 16-agent accounts payable stack. It is ERP-agnostic. Use it on D365, SAP, Oracle, NetSuite, Workday, or a mix. Map field names and transaction codes to your system; do not copy another company's workflow into yours.

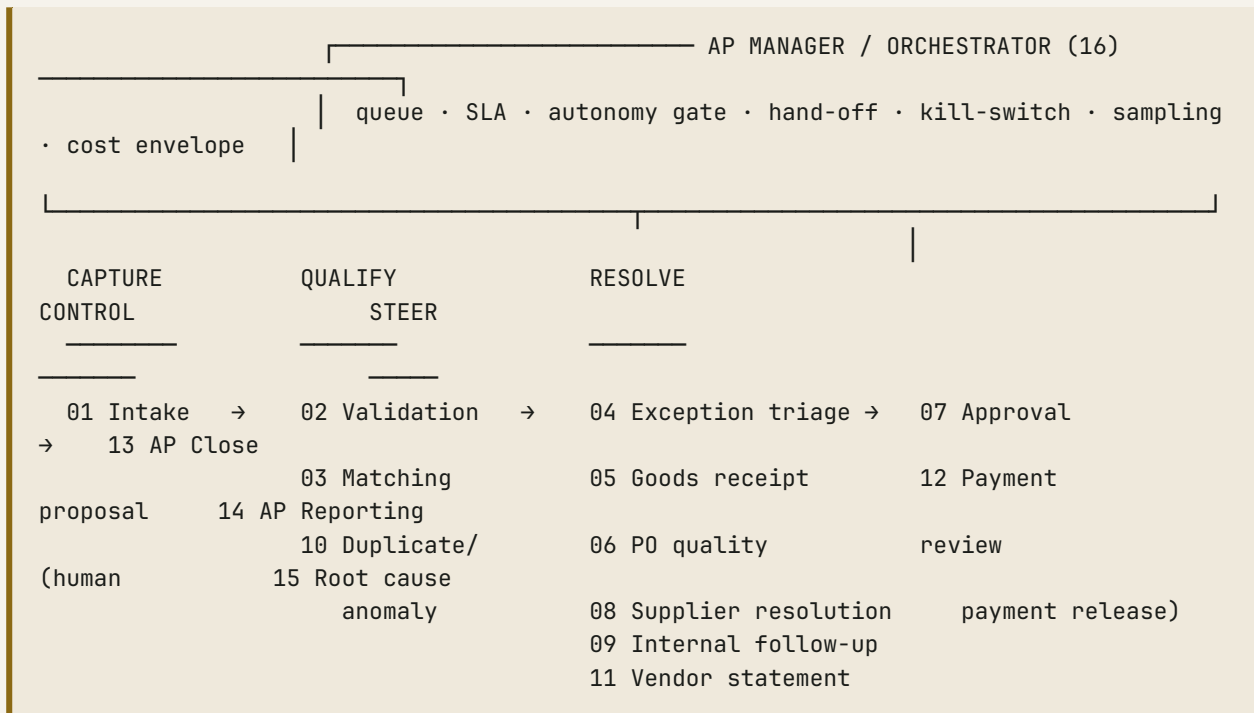
Full autonomy is not the default. An agent starts at **L0 Observe**. It moves only when evidence shows it is safe. The progression is L0 Observe → L1 Recommend → L2 Prepare → L3 Execute within guardrails → L4 Managed autonomy.

What you should do with this library

1. Appoint one **human owner** per agent (typical roles are named in each charter).
2. Run each agent at **L0** on a narrow invoice slice until you can measure precision, miss rate, and exception cost.
3. Promote one level at a time using the gates in `17_AUTONOMY_PROGRESSION.md`.
4. Keep four classes of work **human**: payment release, vendor bank-change approval, policy exceptions, and legal disputes. See `18_HUMAN_VS_AGENT_DECISION.md`.
5. Treat every agent file as a **charter**, not a prompt pack. The instruction skeleton at the end of each file is a starting operating instruction, not a magic prompt.

The 16-agent stack

Work moves left to right. The AP Manager / Orchestrator does not replace specialists. It sequences them, holds the queue, and stops work that lacks evidence.



#	Agent	Job in one line	Typical first owner
01	Invoice Intake	Capture, classify, extract, and park the invoice with an evidence pack	AP Supervisor / Intake Lead
02	Invoice Validation	Prove the invoice is complete, coded, and master-data-fit before match	AP Team Lead
03	Matching	Apply 2-way / 3-way / tolerance / multi-line match and state the residual	AP Processor (PO desk)
04	Exception Triage	Classify the break, name the owner, set the next action and SLA	AP Exception Lead
05	Goods Receipt	Find missing or partial GR evidence; never invent a receipt	Warehouse / Plant AP liaison
06	PO Quality	Find the PO defect that will keep creating exceptions	Procurement Operations
07	Approval	Build the approval packet and route inside published DoA	AP Supervisor
08	Supplier Resolution	Draft the supplier question; send only when the send-gate allows	AP Vendor Desk
09	Internal Follow-Up	Chase the internal owner (receiver, buyer, cost-centre)	AP Query Desk
10	Duplicate & Anomaly	Flag exact and near-duplicate and unusual patterns — not a fraud finding	AP Controls Analyst
11	Vendor Statement Reconciliation	Reconcile the supplier statement to open items	AP Reconciliations
12	Payment Proposal Review	Stress-test the proposal; payment authorisation stays human	Treasury / AP Payments Lead

#	Agent	Job in one line	Typical first owner
13	AP Close	Accruals, GR/IR, cutoff, close checklist	AP Manager / Assistant Controller
14	AP Reporting	Produce the operating pack from defined metrics	AP Manager
15	Root Cause	Turn repeating breaks into a fix request with evidence	Process Owner (AP + Procurement)
16	AP Manager / Orchestrator	Sequence, gate, and stop the stack	AP Manager

How agents hand off

A hand-off is a **structured packet**, not a chat message. The Orchestrator will not accept a packet that is missing required fields.

Packet standard (every hand-off)

Field	Required	Purpose
case_id	Yes	Stable ID for the invoice or statement line across agents
invoice_id / statement_id	Yes	Source document key
legal_entity / company_code	Yes	Which books this hits
vendor_id	Yes, if known	Master-data key; blank is itself a finding
agent_from / agent_to	Yes	Named agents, not people
autonomy_level_applied	Yes	The level used on this case, not the agent's ceiling
decision	Yes	observe / recommend / prepare / execute / hold / escalate
confidence	Yes	0–1, with the rule that produced it
evidence_refs	Yes	Document IDs, ERP transaction IDs, email IDs, hash of source file
exceptions[]	Yes (can be empty)	Code, amount, owner role, SLA clock start
human_required	Yes	Boolean plus reason code
next_action	Yes	One verb, one owner, one due time
cost_envelope	Yes	Inference tokens/cost + expected human minutes if escalated

Default path (happy path)

1. **01 Intake** creates the case, attaches the source file, extracted fields, and file hash. Hands to **02 Validation** and, in parallel, **10 Duplicate & Anomaly**.
2. **10** returns `clear`, `possible_duplicate`, or `anomaly` before **03 Matching** is allowed to execute.
3. **02 Validation** either marks the invoice *match-ready* or hands a coded exception to **04 Triage**.
4. **03 Matching** runs only on match-ready invoices with a PO reference. Non-PO invoices skip 03 and go to **07 Approval** (or coding review) after 02 and 10.
5. Match pass → **07 Approval** if DoA requires it; otherwise to the posted-and-awaiting-payment queue that **12** will later review.
6. Match fail → **04 Triage**, which routes to **05 GR**, **06 PO Quality**, **08 Supplier**, **09 Internal**, or a human queue.

Parallel and return paths

From	To	When
01 Intake	10 Duplicate	Every invoice, before post
03 Matching	05 GR	Qty or receipt variance; GR missing or partial
03 Matching	06 PO Quality	Price, UoM, tax, account-assignment, or vendor-on-PO defect
04 Triage	08 Supplier	Only the supplier can supply the missing fact (credit, revised invoice, POD)
04 Triage	09 Internal	Receiver, buyer, or cost-centre owner must act
11 Statement	01 Intake	Unrecorded invoice found on a statement
11 Statement	04 Triage	Unexplained open item or unapplied credit
12 Payment review	10 Duplicate	Last-look duplicate before the human payment run
12 Payment review	Human payments	Always — agent does not release funds
13 Close / 14 Reporting	15 Root cause	Recurring codes above the review threshold
Any agent	16 Orchestrator	SLA breach, autonomy-gate fail, cost-envelope breach, kill-switch

Hand-off rules that prevent silent failure

- No agent posts, sends, or closes a case without writing `evidence_refs`.
- No agent changes another agent's decision. It can **reject the packet** back to the sender with a reason code.
- **08 Supplier** and **09 Internal** do not invent commercial positions. They transmit a question the triage packet already justified.
- **05 GR** does not create a goods receipt without receipt evidence and a human (or an L3+ gate that still requires evidence).

- **12** may mark a proposal line `hold` , `release-recommended` , or `remove` . A human authorises the run.

What stays human

These four classes are never delegated, at any autonomy level. Agents may prepare the file. They may not decide.

Class	Why it stays human	Agent role
Payment release (bank file, payment run, positive-pay, cheque)	Movement of cash. Irreversible in practice once the bank has it.	Agent 12 reviews and recommends. Human payments / treasury releases.
Vendor bank-change approval (and other payment-master changes)	Classic fraud path. Dual control is the control, not a convenience.	Agent 10 may flag. Agent 08 may collect. A named human approver, not the requester, approves.
Policy exceptions (off-policy spend, after-the-fact PO above threshold, related-party, missing DoA)	Policy is a management decision. An agent cannot grant itself an exception.	Agents 02, 06, 07 detect and package. A named policy owner approves.
Legal disputes (formal claim, solicitor letter, statutory demand, chargeback threat that has left operations)	Legal risk and privilege.	Agents stop operational chasing and hand a file to Legal / AP Manager.

Related human holds (not the four classes, but treated the same in the Orchestrator):

- Write-off or credit acceptance above the published threshold.
- Tax-code change that alters filing position.
- Vendor create / vendor unblock when the vendor is new, high-risk, or previously blocked.
- Any action after the **kill-switch** is on.

How you control this: `human_required=true` is forced by reason code, not by model confidence. Confidence cannot override these classes. See `18_HUMAN_VS_AGENT_DECISION.md` .

The Orchestrator's role (Agent 16)

The AP Manager / Orchestrator is the only agent allowed to:

- Admit a case into the stack and assign the next specialist.
- Read every agent's autonomy ceiling and **downgrade** a case (never silently upgrade).
- Enforce the four human classes and the kill-switch.
- Open or close the cost envelope for a case (max inference spend; max human minutes before a manager sees it).
- Produce the daily operating picture: queue, SLA, first-pass yield, exception mix, cost per invoice.

The Orchestrator does **not**:

- Extract invoices, match lines, write supplier emails, or post journals.
- Approve invoices or release payments.
- Raise its own autonomy. Promotion is a human control change, evidenced in `17_AUTONOMY_PROGRESSION.md`.

If the Orchestrator is down, specialists may continue on already-assigned cases at their current level. New cases park in the intake queue. Payments continue under the existing human payment process.

Autonomy in one page

Level	Name	Agent may	Agent may not	Evidence to promote
L0	Observe	Shadow the process; write what it would have done	Surface work into the live queue	Baseline volume, field-level extraction/match accuracy, miss log
L1	Recommend	Put a recommendation and evidence pack in front of a human	Draft the ERP transaction or the outbound email	Precision / recall on the recommendation; human accept rate
L2	Prepare	Draft the posting, match, email, or GR proposal in a park status	Commit, send, or post	Rework rate on drafts; time-to-approve draft
L3	Execute within guardrails	Commit only when amount, vendor, exception type, and confidence gates pass	Anything outside the published gate	Residual error rate; sample of executed cases; no control-break incidents
L4	Managed autonomy	Execute across the published scope with sampling and a live kill-switch	The four human classes; anything after kill-switch	Sustained L3 evidence plus sampling plan and owner sign-off

Promotion is per **agent × legal entity × invoice slice** (for example: domestic PO invoices under an illustrative threshold). Never promote "the stack."

How you measure the stack

Do not invent a savings number. Measure these, from your own data:

Question	Metric	Formula (use your own counts)
Does capture work?	Intake yield	Invoices with required fields complete / invoices received
Does qualification work?	First-pass yield	Invoices that post or park for payment with no exception / invoices received

Question	Metric	Formula (use your own counts)
Is matching honest?	Touchless match rate	Auto-matched with evidence / PO invoices
Are we faster?	Cycle time	Invoice date (or receipt date — pick one and keep it) → ready-for-payment
Are breaks aging?	Exception aging	Open exceptions by code, owner, and days
Is GR/IR under control?	GR/IR age	Unmatched GR and IR by days and amount
Is the agent cheap enough?	Unit cost	(Inference + tool + human exception minutes × loaded rate) / invoices
Can we prove it?	Evidence completeness	Cases with a complete packet / cases closed

All percentages and amounts in the worked examples in this library are **ILLUSTRATIVE**. Replace them with your ledger.

What can go wrong at stack level

Failure	What it looks like	Control
Silent autonomy upgrade	An agent posts because confidence was high	Orchestrator blocks upgrade; ceiling is a config record, not a model output
Duplicate payment	Same invoice paid twice after a re-extract or a statement load	Agent 10 before post and again before payment; human release
Invented GR	Quantity received to clear a match	Agent 05 requires receipt evidence; L3 still cannot invent
Supplier email that concedes price	Agent 08 "resolves" a variance	Send-gate; commercial concession is a policy exception
Bank-detail change via invoice footer	New IBAN on a PDF treated as an update	Forced human class; Agent 10 flags; master-data dual control
Close without accruals	Unrecorded spend found after period-end	Agent 13 checklist; statement gaps from Agent 11
Cost blow-up	Re-prompting a hard invoice 40 times	Cost envelope per case; Orchestrator stops the case

How you control the stack

- One named **human owner** per agent; one **AP Manager** as Orchestrator owner.
- Autonomy ceiling stored outside the model (config table or control register).
- Kill-switch: AP Manager or Controls can set the stack, an agent, a vendor, or a legal entity to L0 or L1 immediately.

- Sampling: every executed (L3/L4) case is eligible; sample rate is set in the agent's charter.
- Evidence retention: packet + source file + ERP document numbers + who overrode what. Retention follows your finance policy; the library does not set a legal period.
- SoD: the person who approves vendor bank changes is not the person who processes the invoice. The agent does not collapse that.

How to use the files

File	Use it for
AGENT_01 ... AGENT_16	Charters. Implement, train, and audit against these sections.
17_AUTONOMY_PROGRESSION.md	The only promotion method.
18_HUMAN_VS_AGENT_DECISION.md	The decision tree when a case sits on the line.
19_AGENT_CHARTER_STANDARD.md	The template. If you add an agent later, it must pass this standard.

Every page in this library is written to answer at least one of: **What should I do? How? Who owns it? What can go wrong? How do I control it? How do I measure it? What evidence proves it works?**

Monday-morning start (narrow)

Do not stand up sixteen agents.

1. Pick one legal entity and one channel (for example: email PDFs, domestic, PO invoices).
2. Stand up **01 Intake**, **02 Validation**, **03 Matching**, **10 Duplicate**, **16 Orchestrator** at **L0**.
3. Measure field accuracy, duplicate catch, and false match for a period you define.
4. Promote **01** and **10** first — they create the evidence the rest depend on.
5. Leave **12 Payment** at L1 indefinitely until the payments owner asks for L2 prepare-only.
6. Leave bank-change, policy exception, and legal dispute routing as human from day one.

Proof before permission. Agents earn responsibility. Evidence decides.

17 — Autonomy Progression

Product: Evidence Room / AP Agent OS

Line: Agents earn responsibility. Evidence decides.

Owner: AP Manager + Controls

What you should do: Promote one **agent × legal entity × invoice slice** at a time. Never promote "the stack."

Full autonomy is not the default. Every agent starts at **L0 Observe**. A promotion is a control change with evidence, a named approver, a sampling plan, and a way back down.

The five levels

Level	Name	What changes	What does not change
L0	Observe	The agent writes a shadow file: what it would have done, with evidence. Operators do not work from it.	Live ERP, mail, queues
L1	Recommend	Operators see the recommendation and the evidence pack. They decide and they act.	Drafts and posts
L2	Prepare	The agent fills the parked document, draft email, or workflow item. A human submits or sends.	Commit / send / post
L3	Execute within guardrails	The agent commits only when all published gates pass. Everything else stays L2.	Human classes; gates themselves
L4	Managed autonomy	Broader execute inside a published scope, with sampling, a live kill-switch, and a daily exception review.	Human classes; kill-switch authority

L4 is not "unattended AP." It is L3 with a wider slice and a harder operating rhythm. Payment release, vendor bank-change, policy exceptions, and legal disputes **never** enter L3 or L4 decision rights. See `18_HUMAN_VS_AGENT_DECISION.md`.

How you promote

Unit of promotion

```
agent_id × company_code (or legal entity) × slice
```

A **slice** is a documented subset, for example:

- Channel: mailbox PDF / EDI / portal
- Document: PO invoice / non-PO / credit
- Vendor set: named list or "existing vendors only"
- Amount: below a published cap
- Plant or purchasing org

Promoting Agent 03 for domestic stock POs under a cap does **not** promote Agent 03 for service POs.

People

Role	Does
Agent human owner	Proposes promotion; supplies the evidence pack
AP Manager	Accepts or rejects; owns the operating risk

Role	Does
Controls	Confirms the measurement, sampling, and SoD still hold
Table owner (Tax, Procurement, Warehouse, Treasurer)	Signs if their table or risk is in play
Orchestrator config	Updated after signatures; the model cannot update it

Evidence pack (required)

1. **Scope statement** — agent, entity, slice, current level, proposed level.
2. **Volume** — cases in the measurement window (your window; do not invent a standard length).
3. **Accuracy** — the agent's primary KPI vs a human or vs ERP truth, with the method (sample or full).
4. **Misses** — every known miss in the window, with impact (wrong entity, false match, missed duplicate).
5. **False positives / rework** — human reject or edit rate.
6. **Control incidents** — zero expected for promotion; if any, stop.
7. **Cost** — inference + exception minutes.
8. **Gates** that will apply at the new level (amount, vendor, code, confidence, Agent 10 clear, etc.).
9. **Sampling plan** at the new level.
10. **Rollback** — who turns the ceiling down, and the kill-switch test date.

If a number is a target rather than a measured value, label it **TARGET**. If it is an example in this library, it is **ILLUSTRATIVE**. Production packs use your ledger.

Gates to move up

These are **design gates**. Set the numeric thresholds from your own samples. Do not copy illustrative figures into a control policy as if they were norms.

From → to	Minimum evidence
L0 → L1	Shadow file exists; field/decision completeness known; owner can read a miss log; no live-queue use
L1 → L2	Accept rate on recommendations measured; packet standard met; rework reasons classified; draft API in a park status only
L2 → L3	Rework rate measured; residual error on a sample of what <i>would</i> have executed; gates written as config (not prompt text); Agent 10 execute-ready if the agent can post; kill-switch drill done; SoD reviewed
L3 → L4	Sustained L3 window with sample results; no control incidents; sampling plan for the wider slice; daily review owner named; Treasurer sign-off if the agent touches payment proposals (holds only)

Refuse promotion if: Agent 10 cannot run; packet `evidence_refs` are incomplete; the owner wants to "skip L2"; the slice is "all invoices"; a human class is in the proposed execute set.

How you downgrade and stop

Trigger	Action
Control incident (wrong post, supplier concession sent, GR without evidence, near-duplicate paid)	Immediate ceiling to L1 or L0 for that slice; incident review
Kill-switch	Orchestrator propagates; specialists behave as L0/L1 as set
Sample fail	Ceiling -1 until a new pack
Cost-envelope thrash	Stop the case; if clustered, ceiling -1
Owner absence (no reviewer for L2 drafts)	Do not "compensate" with L3; stay L1 or pause the slice
Agent 10 down	Execute-level 02/03/12 stop (fail closed)

Downgrade does not need the full promotion pack. It needs a reason code and a config write. Raising again needs a new pack.

Sampling (L3 and L4)

- Every executed case is **eligible**.
- Sample rate is set per agent charter (higher for GR posts, payment holds, first weeks of L3).
- Sampler is a human who did not clear the same case.
- Findings: `agree` / `agree-with-edit` / `disagree-control` / `disagree-quality`.
- `disagree-control` is an incident.

Sampling is how you prove L3 works. Volume of execution is not proof.

What "confidence" is allowed to do

Confidence may **stop** an execute (fall back to L2).

Confidence may **not** widen a tolerance, skip Agent 10, skip DoA, or clear a human-class reason code.

Publish how confidence is computed (rule-pass count, extractor scores). If you cannot explain it, do not gate on it — gate on rules only.

Suggested first promotions (not a mandate)

A safe order for a first entity:

1. **01 Intake** L0→L1→L2 (park cases).

- 2. **10 Duplicate** L0→L1→L2 holds on exact rules.
- 3. **16 Orchestrator** L1→L2 assignment.
- 4. **02 Validation** L1→L2 on PO invoices.
- 5. **03 Matching** L1→L2 park on 3-way stock POs.
- 6. **04 Triage** L1→L2 proposed queues.
- 7. Then **09** and **08** at L2 (send-gate on).
- 8. **12** stays L1 (or L2 prepare) until the Payments Lead asks. L3 holds are optional and late.
- 9. **05 / 06 / 07 / 11 / 13 / 14 / 15** follow their charters; several should stay at L1/L2.

Do not L3 Agent 05 (GR post) to "help" Agent 03. That hides receiving problems and creates inventory noise.

Config record (store outside the model)

Field	Example shape
agent_id	03_matching
legal_entity	DE01
slice_id	stock-po-3way-mailbox
ceiling	L2
gates	amount cap, vendor set, codes allowed
effective_from	timestamp
approved_by	user IDs
evidence_pack_uri	
sample_rate	
kill_switch	off

The Orchestrator reads this. Specialists do not write it.

Worked example — ACME Manufacturing (fictional)

ACME (fictional) wants Agent 03 at L3 for DE01 , stock POs, 3-way, mailbox, existing vendors, gross below a published cap.

They may promote when (shape, not a norm): a measured L2 window shows park drafts the PO desk accepts; a sample of "would have executed" cases has no false match; tolerances are the ERP table, not a new table; Agent 10 exact rules are on; kill-switch drill completed; Controls signed.

They stay at L2 if service POs, multi-PO invoices, or incidental freight are still in the same slice — split the slice first.

They roll back if a sampled L3 post used a reversed GR or converted 3-way to 2-way because GR was missing.

How you measure progression

Question	Metric
Are we earning it?	Packs approved / packs proposed
Are we honest?	Downgrades and kill-switch events
Are we sampling?	Sampled executes / executes
Are we leaking?	Executes above ceiling (target zero)

What can go wrong

- Silent upgrade inside a prompt ("just post it").
- One ceiling for all entities.
- L3 on Friday before a payment run without a sample.
- Promoting 08 send-gate off to "clear the queue."
- Calling L4 "autonomous AP" in a steering pack.

How you control it

- Config outside the model.
- Two signatures (AP Manager + Controls) for L2→L3 and L3→L4.
- Kill-switch operable if the model is down.
- This document is the only promotion method.

Proof before permission.

18 — Human vs Agent Decision

Product: Evidence Room / AP Agent OS

Owner: AP Manager + Controls

What you should do: When a case sits on the line, use this tree. Confidence is not a vote. Human classes cannot be cleared by the model.

The rule

Agents **prepare and, when earned, execute inside gates.**

Humans **authorise cash, master payment data, policy exceptions, and legal positions.**

If you are unsure, the case is human. Fail closed.

Four classes that stay human at every level

Class	Examples	Agent may	Human must
Payment release	Payment run approve, bank file, BCM, positive-pay, cheque release, urgent outbound	Agent 12: review, hold, remove, recommend	Authorise the run and the file
Vendor bank-change	IBAN, account, intermediary, factoring payee, payee name	Flag (ANOM-PAYEE-DETAIL); collect documents	Dual-control master-data approve. Not the invoice processor alone
Policy exception	Missing PO on PO-required spend; after-the-fact PO above threshold; related-party; off-policy vendor; DoA skip	Detect, package the form	Named policy owner
Legal dispute	Solicitor letter, statutory demand, formal claim, collection that left operations	Stop 08/09 operational chase; file the packet	Legal / AP Manager

Related holds treated the same:

- Write-off / credit acceptance above threshold.
- Tax treatment that changes filing position.
- Vendor create / unblock when new, high-risk, or previously blocked.
- Any action after kill-switch.

`human_required=true` is forced by **reason code**. A high confidence score cannot flip it to false.

Decision tree

Walk top to bottom. First match wins.

1. Kill-switch on for stack / agent / vendor / entity / channel?
→ Human (or L0 only). Orchestrator stops execute.
2. Is this payment release or bank-file send?
→ Human. Agent 12 pack only.
3. Is this vendor payment-master change, or printed payee ≠ master?
→ Human dual control. Agent 10 flags. Do not update from the invoice.
4. Legal / solicitor / statutory demand / threat that left operations?
→ Human Legal. Stop 08/09.
5. Policy exception (P0-required missing, confirming P0 over threshold, related-party, DoA skip, off-policy)?
→ Human policy owner. Agents package.
6. Write-off, tax-position change, vendor unblock/create in the high-risk set?
→ Human per table.
7. Agent 10 not clear (possible_duplicate, anomaly uncleared, index down)?
→ No execute on 02/03/12. Human or wait.
8. Packet missing evidence_refs or packet standard fail?
→ Reject to sender. No execute.
9. Case cost envelope exhausted?
→ Stop. Human or next-day replay.
10. Proposed action is inside the agent's published L3/L4 gate (amount, vendor set, code, template, evidence type)?
→ Agent may execute at ceiling.
Else → Agent prepares (L2) or recommends (L1). Human acts.

By agent — default human line

Agent	Agent side of the line	Human side of the line
01 Intake	Extract, store, split, propose entity/ vendor	New vendor; legal mail; apply printed bank details
02 Validation	Completeness, arithmetic, coding proposal	Grant missing-PO; clear vendor block; tax- position change
03 Matching	Compare using published tolerances	Post outside tolerance; convert 3-way to 2-way; create GR
04 Triage	Route, bundle, SLA	Close without resolution; policy close; legal route
05 GR	Find evidence; prepare GR	Post without non-invoice evidence; invent qty; QM override
06 PO Quality	Diagnose; clerical allow-list if published	Price, vendor, qty increase, confirming PO

Agent	Agent side of the line	Human side of the line
07 Approval	Route, remind, apply published skip	Be the approver; invent delegate
08 Supplier	Template draft; send only if send-gate allows	Send-gate default; concession; bank change; legal reply
09 Internal	Template chase to resolved employees	Chat-as-posting; external domains; director escalate off-table
10 Dup/ Anomaly	Flag, exact hold	Fraud opinion; vendor block; release DUP-PAID without reason
11 Statement	Recon pack, tickets	Pay statement total; invoice from line (default no)
12 Pay review	Hold/remove/recommend	Release run; send file; change payee
13 Close	Checklist, GR-backed candidates, park journals	Post (default); period status; statement-gap accrual
14 Reporting	Dictionary metrics	Savings claims; external publish; drop a red metric
15 Root cause	Fix request	Change tables; weaken 3-way/DoA/bank control
16 Orchestrator	Assign, stop, propagate kill-switch	Set kill-switch; raise ceilings; specialist work

"The supplier will cut us off" and other pressure

Operational pressure does not move a case across the line.

Pressure	Still human	Agent help
Pay today or no next shipment	Payment release	12 pack + AP Manager
"Here is our new bank account" on PDF or email	Bank-change dual control	10 flag; 08 stop
Buyer says "just raise the PO" after the invoice	Confirming PO / policy	06 pack
Warehouse says "I received it" in chat	Not a GR	09 testimony → 05 still needs a document
Controller wants close now	Period status	13 pack as-is, incomplete marked

Overrides

An override is a human clearing a hold or executing outside a gate.

Rule	Detail
Who	Named role in the agent charter; not the agent

Rule	Detail
Dual control	Required for DUP-PAID , PAY-PAYEE-MISMATCH , bank-change, write-off above threshold
Reason code	Mandatory
Logging	Old decision, new decision, user, time
Same-person ban	Override ≠ original vendor create ≠ payment release

Repeated overrides of an exact duplicate rule mean the **normaliser is wrong**, not that the rule should go away.

How you measure the split

Metric	Formula
Human-class leakage	Execute events on human-class reason codes (target zero)
Override rate	Overrides / holds
Dual-control breaches	Single-person bank or payee overrides (target zero)
Fail-closed events	Stops because 10/index/envelope — count them; they are working

Worked example — ACME Manufacturing (fictional)

ACME (fictional) steel supplier emails: "Pay invoice 4500123 to IBAN DE89... (new). Production stops Friday." Invoice already match-ready after the GR. Proposal is tomorrow.

Tree.

1. Kill-switch off.
2. Paying is payment release → human (12 + treasurer).
3. New IBAN → bank-change human class. Agent 10 ANOM-PAYEE-DETAIL . Agent 08 does not reply "we will update."
4. Agent 12 holds the line PAY-PAYEE-MISMATCH even if the goods are real.
5. Vendor-master dual control runs **separately**. If they approve the IBAN, the **next** proposal may include it. This run does not.
6. AP Manager talks to the plant about supply risk. That conversation is not an agent decision.

Wrong outcome. Agent 12 release-recommended because the match was clean. Or Agent 08 accepts the IBAN.

What can go wrong

- Treating "urgent" as a skip.
- Using model confidence as a second signature.
- Letting 08 send from a new email on the invoice.
- Letting 05 post GR from the invoice to protect a supplier relationship.
- Writing "agent approved payment" in ERP text.

How you control it

- Reason codes in config.
- Orchestrator cannot clear them.
- Charters in this library state the line per agent.
- Sampling includes "was this a human class?"

Evidence decides. Permission follows.

Evidence Room — AP Agent OS Professional

Process Mapping — 00 Methodology

Product: Evidence Room AP Agent OS — Professional

Module: Process Mapping

Standard: Proof before permission

Audience: Head of AP, Shared Services lead, Transformation lead, Process Excellence, Internal Audit

ERP stance: Agnostic. Record the buyer's system names in the process register; do not assume a vendor.

Examples: ACME Industrial Holdings — **ILLUSTRATIVE** only.

Version: 1.0

Classification: Internal operating standard. Buyer completes placeholders marked [BUYER].

How to use this document

This is the ten-step Evidence Room method for converting a live Accounts Payable operation into a governed agent workflow. Use it as the master sequence. The other files in this folder are the working instruments:

Step	Template / instrument
Observe	01_PROCESS_DISCOVERY_GUIDE.md
Transcribe	02_TRANSCRIPT_TO_WORKFLOW.md

Step	Template / instrument
Extract / Structure / SOP	03_SOP_GENERATOR_FRAMEWORK.md
RACI	04_RACI_STANDARD.md
Agentise / autonomy	../Governance/01_AUTONOMY_POLICY.md
Test / Shadow / Pilot	../Governance/02_RELEASE_AND_CHANGE.md and ../Testing/
Measure	../KPI_and_Measurement/
Expand responsibility	../Governance/01_AUTONOMY_POLICY.md

Do not skip steps to “get to the agent faster.” Permission to act is earned by evidence produced in earlier steps.

Operating rule for every step

Each step below answers the same seven questions:

1. **What to do** — the work itself.
2. **How** — the method.
3. **Who** — named role, not “the team.”
4. **What can go wrong** — failure modes that invalidate later steps.
5. **Control** — the check that the step was done properly.
6. **Measure** — the evidence that the step completed.
7. **Evidence** — the artefact stored in the Evidence Room.

Timeboxes are **ILLUSTRATIVE** planning ranges. Replace them with the buyer’s calendar.

The ten-step method

Observe → Transcribe → Extract → Structure → Agentise
→ Test → Shadow → Controlled Pilot → Measure → Expand responsibility

Responsibility expands only after Measure produces a signed evidence pack. If Measure fails, return to Test or Agentise. Do not jump to Expand.

STEP 1 — Observe

Purpose. Record how invoices, exceptions, approvals and payments actually move today — not how the SOP says they move.

What to do. Sit with AP clerks, exception handlers, buyers, goods-receipt owners and the payment preparer. Watch a complete cycle: intake → validation → match/exception → approval → payment proposal → close items.

How. Use 01_PROCESS_DISCOVERY_GUIDE.md . Run at least one live walkthrough per material channel (email, portal, EDI, scan). Photograph or export screens only where policy allows. Timestamp every observation.

Who.

Role	Duty
Transformation / Process Excellence lead	Designs the observation plan and timebox
AP process owner	Grants access and names the observed operators
Observed operator	Works normally; does not “perform the SOP”
Internal Audit (optional observer)	Confirms control points are not staged

Inputs. Current SOP (if any), org chart, system access list, sample invoice IDs, exception queue export, payment calendar.

Outputs. Observation log; channel inventory; system/touch inventory; “as-done vs as-written” gap list.

Timebox. ILLUSTRATIVE: 3–5 working days for a single legal entity and one AP team. Multi-entity shared services: one additional day per material entity.

Failure modes.

- Operators demonstrate the happy path only.
- Observer records system clicks but misses informal Slack/email workarounds.
- Sample is all PO-matched invoices; non-PO and intercompany are skipped.
- Observation is of UAT, not production.

Template used. Observation log and interview pack in 01_PROCESS_DISCOVERY_GUIDE.md .

Control. Process owner signs that the observed sample covers the agreed channels and exception types.

Measure. Count of observed invoices, exception types seen, systems touched, informal workarounds noted.

Evidence. Signed observation log stored under [BUYER]/Evidence/Observe/ .

ACME — ILLUSTRATIVE. Observer watches 12 invoices: 6 PO-match, 3 missing GR, 2 non-PO, 1 duplicate suspect. Discovers clerks paste VAT IDs from a desktop spreadsheet that is not in the SOP.

STEP 2 — Transcribe

Purpose. Produce a durable, reviewable record of how people describe the work, the rules they apply, and the exceptions they recognise.

What to do. Transcribe walkthroughs, exception huddles and payment-prep meetings. Keep speaker labels and timestamps.

How. Use 02_TRANSCRIPT_TO_WORKFLOW.md . Prefer recorded sessions with consent. If recording is refused, produce a contemporaneous structured note using the same headings. Do not polish language; preserve decision language (“we usually...”, “if it’s over...”, “I ask Jane”).

Who.

Role	Duty
Scribe / analyst	Produces the transcript and speaker map
Session owner (AP Manager)	Confirms attendees and consent
Legal / HR (if required)	Confirms recording is permitted

Inputs. Observation schedule, consent record, audio/video or live notes, speaker roster.

Outputs. Timestamped transcript; speaker map; rule-candidate list (raw); unresolved questions.

Timebox. **ILLUSTRATIVE:** same-day draft; review within 2 working days of the session.

Failure modes.

- Transcript merges speakers; rules cannot be attributed.
- “Usually” and “always” are treated as the same.
- Confidential supplier or payroll content is stored without redaction.
- Transcript is rewritten into a clean SOP before Extract, destroying evidence of variation.

Template used. Transcript header, speaker map and redaction rules in 02_TRANSCRIPT_TO_WORKFLOW.md .

Control. Session owner initials the speaker map. Privacy owner confirms redaction of personal and banking data.

Measure. Hours recorded vs hours transcribed; open questions remaining; redaction completeness check.

Evidence. Transcript file + consent + redaction log under [BUYER]/Evidence/Transcribe/ .

STEP 3 — Extract

Purpose. Pull discrete, testable objects from the transcript and observation log. No design yet.

What to do. Identify steps, systems, decisions, business rules, inputs, outputs, exceptions, controls and dependencies.

How. Tag the transcript with the object types below. One object per row. Quote the source line. Do not invent a rule that nobody stated.

Object	Extract as
Step	Verb + object + system (“park invoice in ERP”)
System	Named application or mailbox, not “the system”

Object	Extract as
Decision	Condition + outcomes
Business rule	Threshold, tolerance, policy clause, or local habit
Input	Document, field, or data feed required to start the step
Output	Artefact the step produces
Exception	Condition that stops STP, mapped later to 03_EXCEPTION_TAXONOMY.md
Control	Check that is supposed to prevent or detect error
Dependency	Person, team, or system the step waits on

Who. Analyst extracts; AP process owner reviews object list; control owner reviews control objects.

Inputs. Transcript, observation log, sample documents, current policy excerpts.

Outputs. Object register (one row per object, with source quote and timestamp).

Timebox. **ILLUSTRATIVE:** 1–2 days per recorded session.

Failure modes.

- Analyst “corrects” a stated rule to match policy.
- Informal rules are discarded as “not real.”
- Exception names do not match the taxonomy codes.
- Controls are listed without saying who performs them.

Template used. Extract register in 02_TRANSCRIPT_TO_WORKFLOW.md (Extract section).

Control. Dual review: process owner for operational objects; control owner for control objects. Unquoted objects are rejected.

Measure. Objects extracted; objects rejected for missing source; % of exception objects mapped to a taxonomy code.

Evidence. Object register under [BUYER]/Evidence/Extract/.

ACME — ILLUSTRATIVE. Transcript line 00:41:12: “If the price is more than a couple of percent I send it back to the buyer.” Extracted rule: price-tolerance exists but is not quantified. Flagged as an open policy gap, not written as “2%.”

STEP 4 — Structure

Purpose. Convert extracted objects into artefacts a second person can operate, test and audit.

What to do. Produce a process map, decision tree, exception map, control map, RACI and draft SOP.

How. Use 03_SOP_GENERATOR_FRAMEWORK.md and 04_RACI_STANDARD.md . Exception codes must come from ../Controls/03_EXCEPTION_TAXONOMY.md . Do not create local exception names.

Who.

Artefact	Author	Approver
Process map	Analyst	AP process owner
Decision tree	Analyst	Policy owner (Procurement / Tax / Treasury as relevant)
Exception map	Exception lead	AP Manager
Control map	Control owner	Internal Audit (review)
RACI	AP Manager	Finance Controller
SOP	Process owner	Controller / Head of AP

Inputs. Object register; taxonomy; current DOA; current SoD matrix.

Outputs. Versioned process pack: map, trees, exception map, control map, RACI, SOP draft.

Timebox. ILLUSTRATIVE: 5–8 working days for one end-to-end AP path (intake to payment proposal).

Failure modes.

- Process map shows the target state disguised as current state.
- RACI assigns an agent as Accountable. Agents are never Accountable.
- SOP includes “use judgement” with no decision tree.
- Exception map uses free-text reasons instead of taxonomy codes.

Template used. 03_SOP_GENERATOR_FRAMEWORK.md , 04_RACI_STANDARD.md , ../Controls/03_EXCEPTION_TAXONOMY.md .

Control. Pack checklist signed: current-state labelled as current-state; every exception has a code; every control has an owner.

Measure. Open policy gaps remaining; unmapped exceptions; RACI conflicts (two Accountables, or none).

Evidence. Versioned process pack under [BUYER]/Evidence/Structure/ .

STEP 5 — Agentise

Purpose. Decide, per step, what remains human, what an agent may recommend, prepare or execute, and where a deterministic rule is preferable to a model.

What to do. Score each structured step against the autonomy ladder in ../Governance/01_AUTONOMY_POLICY.md . Default every new agent to Level 0 (Observe) or Level 1 (Recommend). Execution is not a design-time gift.

How. For each step record:

Field	Rule
Human-retained	Irreversible money movement, bank-detail change, policy exception above DOA, override of a control
Deterministic automation preferred	Exact field compares, tolerance maths, duplicate exact-key match, calendar ageing
Model / agent candidate	Classification, draft communication, root-cause clustering, incomplete-document review
Proposed autonomy	L0–L4, with required evidence to promote
Explicit exclusions	What the agent must never do

Who. AP Manager proposes; Control owner and SoD owner challenge; Head of AP / Controller approves the agent charter. IT / security reviews tool access.

Inputs. Structured pack; risk register; DOA; data-access inventory.

Outputs. Agent charter(s); human/agent/deterministic split; tool and data list; exclusions; proposed test set.

Timebox. ILLUSTRATIVE: 3–5 working days per agent after the structured pack is signed.

Failure modes.

- Designer grants Level 3 because “the use case is simple.”
- LLM is used for exact PO-price compare.
- Agent is given ERP posting rights “for the pilot.”
- Exclusions are omitted; the charter is therefore untestable.

Template used. Agent charter fields in `../Governance/01_AUTONOMY_POLICY.md` ; control rows in `../Controls/01_CONTROL_MATRIX.md` .

Control. Charter cannot be approved if Accountable is an agent, if payment execution is in scope, or if exclusions are blank.

Measure. Steps classified; % deterministic vs model; number of L2+ proposals (should be zero at first design unless evidence already exists).

Evidence. Signed agent charter under `[BUYER]/Evidence/Agentise/` .

ACME — ILLUSTRATIVE. Matching Agent: price/qty/receipt compare = deterministic. Exception classification = agent at L1. Supplier email draft = L2 (prepare, human send). ERP park/post = out of scope.

STEP 6 — Test

Purpose. Prove the agent and the structured rules against historical cases before they see live work.

What to do. Build a labelled case file. Run the agent and the deterministic rules. Score accuracy, false positives, false negatives, and handling of out-of-scope items.

How. Pull a stratified historical sample: STP, each material exception code, high value, intercompany, credit notes, foreign currency if used. Humans label the expected outcome first. Then run the agent. Disagreements are adjudicated by the process owner, not the model vendor.

Who. Analyst builds the file; AP specialists label; Control owner reviews scoring rules; IT provisions a non-production or read-only environment.

Inputs. Historical invoice IDs; taxonomy; agent charter; scoring definitions from `../KPI_and_Measurement/01_KPI_DICTIONARY.md`.

Outputs. Test report: n, accuracy, FPR, FNR, residual error list, out-of-scope handling, go/no-go.

Timebox. ILLUSTRATIVE: 5–10 working days including labelling. Do not start Shadow until the agreed gates are met.

Failure modes.

- Sample is only recent easy invoices.
- Labels are created after seeing the agent output.
- Production credentials are used.
- “Close enough” is scored as correct.

Template used. Testing pack in `../Testing/` (where present) and KPI dictionary definitions.

Control. Labelling precedes model output. Environment is read-only or non-production. Scorecard uses dictionary formulas, not vendor dashboards.

Measure. Sample size by exception code; accuracy; FPR; FNR; residual high-risk errors.

Evidence. Test report + labelled file IDs under `[BUYER]/Evidence/Test/`.

Gates are buyer-set. **Do not treat any percentage in an ACME example as a target.**

STEP 7 — Shadow

Purpose. Run the agent on live work with **no action permissions**. Compare agent output to human output in real time.

What to do. Agent reads the same queue the humans work. It writes recommendations to the Evidence Room only. Humans work as today. Daily reconciliation of agree / disagree / not-comparable.

How. Autonomy locked at L0 or L1 with write-block. No email send, no ERP change, no workflow status change. Shadow period covers a full payment cycle if the agent touches payment-adjacent work.

Who. AP Manager runs the shadow; operators continue their jobs; Control owner reviews disagreements that touch SoD, bank data, or high-value items.

Inputs. Live queue; agent charter; test-gate evidence; access locked to read + write-to-evidence-only.

Outputs. Shadow log; disagreement file; operator feedback; promote / hold / redesign recommendation.

Timebox. ILLUSTRATIVE: one full AP cycle (often 2–4 weeks). Extend if volume is too low to see material exception codes.

Failure modes.

- Agent silently gains a “send” or “post” permission.
- Humans start following the agent before promotion.
- Shadow covers only quiet weeks.
- Disagreements are discarded as “user error” without review.

Template used. Shadow log fields in `../Governance/01_AUTONOMY_POLICY.md` and weekly report in `../KPI_and_Measurement/03_WEEKLY_AGENT_REPORT.md`.

Control. Access review at start and mid-point: agent identities have no execution entitlements. Daily shadow log signed by AP Manager.

Measure. Agreement rate; material disagreements; exception codes not seen; operator minutes spent reviewing the agent (should be recorded, not assumed to be “free”).

Evidence. Shadow log + access review under `[BUYER]/Evidence/Shadow/`.

STEP 8 — Controlled Pilot

Purpose. Allow a narrow, reversible set of agent actions under human ownership.

What to do. Promote only the actions that Shadow supported. Limit by user, legal entity, supplier segment, invoice type, value cap and calendar window.

How. Follow `../Governance/02_RELEASE_AND_CHANGE.md`. Pilot scope is written as allow-lists, not deny-lists. Fallback and override are live before the first pilot transaction (`../Governance/03_INCIDENT_AND_OVERRIDE.md`).

Who. Head of AP / Controller is Accountable for the pilot. AP Manager is Responsible for daily operation. Control owner and IT are Consulted. Internal Audit is Informed.

Inputs. Shadow evidence pack; signed pilot charter; rollback plan; communication to affected suppliers/internal users if they will receive agent-drafted mail.

Outputs. Pilot charter; daily exception of pilot vs non-pilot; incident log; mid-pilot review; exit report.

Timebox. ILLUSTRATIVE: 4–8 weeks. Do not expand scope mid-pilot because “it looks fine.”

Failure modes.

- Scope creeps to all suppliers in week two.

- Override is used as the normal path.
- Pilot success is declared on activity ("emails sent") not outcomes.
- Rollback is untested.

Template used. Release record in `../Governance/02_RELEASE_AND_CHANGE.md` ; incident/override in `../Governance/03_INCIDENT_AND_OVERRIDE.md` .

Control. Change Advisory (or Finance Systems equivalent) records the pilot as a change. Value cap and entity allow-list are technically enforced, not "agreed in a meeting."

Measure. Pilot volume; override rate; incidents; outcome KPIs vs baseline for the same segment.

Evidence. Pilot charter + mid-point review + exit report under `[BUYER]/Evidence/Pilot/` .

ACME — ILLUSTRATIVE. Goods Receipt Agent drafts chasers only for domestic inventory POs under a buyer-set value cap, one plant, two named requesters copied. No chasers to Directors. No ERP receipt creation.

STEP 9 — Measure

Purpose. Compare pilot (or shadow, if still pre-action) against the pre-agreed baseline using the KPI dictionary. Decide with numbers the buyer owns.

What to do. Score Activity, Operational, Financial and Risk-control outcomes separately. Do not offset a control breach with a cycle-time improvement.

How. Use `../KPI_and_Measurement/00_KPI_FRAMEWORK.md` and `01_KPI_DICTIONARY.md` . Baseline is frozen before Pilot. Savings are recorded only when validated by Finance; inferred "hours released" stay in a separate, labelled line.

Who. AP Reporting owner compiles; AP Manager explains variances; Controller / Head of AP decides; Internal Audit may review the pack.

Inputs. Baseline worksheet; dictionary; agent cost ledger; incident/override log; sample QA file.

Outputs. Measurement pack; promote / hold / roll back recommendation; updated risk register entries.

Timebox. ILLUSTRATIVE: 5 working days after pilot window closes, plus however long Finance needs to validate any claimed cash effect.

Failure modes.

- Vendor dashboard replaces dictionary definitions.
- Baseline was never frozen.
- Financial lines include unvalidated "opportunity."
- Small-n exception codes are reported as rates without a count.

Template used. `../KPI_and_Measurement/02_SCORECARD_GUIDE.md` , `03_WEEKLY_AGENT_REPORT.md` , `04_MANAGEMENT_DASHBOARD.md` .

Control. Every reported KPI cites its dictionary ID. Financial lines require a Finance sign-off field. Risk-control lines cannot be hidden.

Measure. Completeness of the scorecard; number of KPIs with n below the buyer's minimum; unvalidated financial claims (must be zero on the signed pack).

Evidence. Signed measurement pack under [BUYER]/Evidence/Measure/.

STEP 10 — Expand responsibility

Purpose. Increase autonomy or scope only when the measurement pack shows the agent earned it.

What to do. Apply the promotion rules in `../Governance/01_AUTONOMY_POLICY.md`. Expansion is a change, not a setting toggle. Each expansion has its own Test → Shadow (if scope is new) → Pilot → Measure cycle, sized to the increment.

How. Allowed increments (one at a time unless the change board records a reason):

1. Autonomy +1 level, same scope.
2. Same autonomy, one added legal entity or channel.
3. Same autonomy, one added exception code.
4. Same autonomy, higher value cap.

Never combine "higher autonomy + broader scope + higher value" in one change.

Who. AP Manager proposes; Control owner and SoD owner must approve; Head of AP / Controller is Accountable; Change record required.

Inputs. Measurement pack; residual error list; incident history; current entitlements; updated charter.

Outputs. New charter version; change record; communication; next evidence plan.

Timebox. ILLUSTRATIVE: decision within 10 working days of a complete measurement pack. Implementation time is a separate change window.

Failure modes.

- Promotion because a senior stakeholder "wants to see AI working."
- Scope and autonomy raised together.
- Previous incidents closed as "user training" without a control change.
- Model or prompt changed in the same release as a promotion.

Template used. Autonomy policy; release and change; control matrix (updated rows).

Control. Promotion checklist: measurement pack complete; no open Critical incidents; entitlements match the new level; model/prompt hash unchanged unless separately tested.

Measure. Autonomy level before/after; scope delta; time since last material incident; residual FNR on high-risk codes.

Evidence. Signed promotion record under [BUYER]/Evidence/Expand/ .

Sequence controls (the method as a control)

Gate	Cannot start until	Evidence
Transcribe	Observation plan signed; consent in place	Observe pack
Extract	Transcript accepted; redaction complete	Transcribe pack
Structure	Object register dual-reviewed	Extract pack
Agentise	Current-state pack signed (not target-state)	Structure pack
Test	Charter signed; environment confirmed non-executing	Agentise pack
Shadow	Test go/no-go passed; access review	Test pack
Controlled Pilot	Shadow pack; rollback tested; allow-lists enforced	Shadow + release record
Measure	Baseline frozen before pilot start	Pilot data + baseline
Expand	Measurement pack signed; one-increment rule	Measure pack

If a gate artefact is missing, the next step is unauthorised — even if the software is technically able to run.

Roles that persist across all ten steps

Role	Accountability
Head of AP / Controller	Accountable for the operating model and any agent that touches AP work
AP Manager	Responsible for daily method discipline and agent performance
Control / Risk owner	Challenges autonomy, SoD, and evidence quality
Finance Systems / IT	Access, logging, environments, model/vendor change
Internal Audit	Periodic inspection of the Evidence Room, not day-to-day operation
Agent (any)	Never Accountable. Never the owner of a control.

What this method is not

- A promise that agents will reduce cost or cycle time.
- A compliance certification.
- A licence to bypass ERP controls.
- A substitute for DOA, SoD, or payment authorisation.

Proof before permission.

End of 00_METHODOLOGY.md

Evidence Room — AP Agent OS Professional

Governance — 00 Framework

Product: Evidence Room AP Agent OS — Professional
Module: Governance
Standard: Proof before permission
Audience: CFO, Controller, Head of AP, Control / Risk, Internal Audit, Finance Systems, Privacy, Legal
ERP stance: Agnostic. Governance binds people, identities and evidence — not a vendor module.
Examples: ACME Industrial Holdings — **ILLUSTRATIVE** only.
Version: 1.0
Classification: Binding operating standard. Buyer completes [BUYER] fields. This document does **not** certify legal, regulatory, accounting, or security compliance.

Position

Agents are tools with job descriptions. They earn scope. They never hold accountability. This framework is the set of rules that makes that sentence operational.

Use with:

Topic	File
Autonomy ladder and promotion	01_AUTONOMY_POLICY.md
Testing, release, model and workflow change	02_RELEASE_AND_CHANGE.md
Incidents, override, fallback, BCP	03_INCIDENT_AND_OVERRIDE.md
Control design and matrix	../Controls/
Measurement	../KPI_and_Measurement/
Who is A/R/C/I	../Process_Mapping/04_RACI_STANDARD.md

Every section below uses the same seven answers: what to do / how / who / what can go wrong / control / measure / evidence.

1. Human accountability

What to do. Name one human Accountable for every agent, every control, every exception family, and every release.

How. Charter field `Accountable` is a standard role plus named incumbent (`04_RACI_STANDARD.md`). Deputies are named. Absence of the incumbent does not transfer accountability to the agent or to a vendor.

Who. Head of AP / Controller for operating agents; Controller for financial claims; CFO / Finance Director for L3+ autonomy and any payment-adjacent expansion.

What can go wrong. "The model did it." Vendor success-manager treated as owner. Shared mailbox as Accountable.

Control. Charter gate rejects blank, vendor, mailbox, or agent Accountable. Quarterly incumbency certification.

Measure. Charters with valid human A; recertifications overdue.

Evidence. Charter register; certification file.

2. Segregation of duties (SoD)

What to do. Keep incompatible duties apart for humans **and** for the system identities agents use.

How. Default incompatible pairs (extend locally):

Duty A	Duty B
Invoice capture / entry	Payment authorisation
Vendor master create	Payment authorisation
Vendor bank-detail change	Payment authorisation
Invoice approval (DOA)	Payment authorisation (unless local policy explicitly combines — treat as residual risk)
Exception close above [BUYER cap]	Same person who created the invoice record
Agent entitlement grant	Agent performance accept (Measure)

Agent identities inherit the narrower of: (a) the human role they assist, (b) the exclusions on the charter. An orchestrator must not hold payment-release or master-data-write entitlements.

Who. Control owner maintains the pair list. Finance Systems implements. Controller approves exceptions.

What can go wrong. A "service account" that can park **and** propose **and** release. An agent that changes vendor bank data because the connected API token was reused.

Control. Access review at every promotion and quarterly. SoD exceptions have risk-register ID and expiry.

Measure. Open SoD exceptions; identities with incompatible entitlements (count must be explainable).

Evidence. Access extract; exception file; `../Controls/01_CONTROL_MATRIX.md` rows.

3. Least privilege

What to do. Give each agent identity the minimum objects, verbs and records needed for the current autonomy level and allow-list.

How. Privilege is an allow-list: entities, channels, supplier segments, exception codes, value cap, verbs (`read` , `write-evidence` , `draft` , `send` , `park` , `post` — `post` is default **denied**). Shadow = `read` + `write-evidence` only. Promotion adds one increment (`01_AUTONOMY_POLICY.md`).

Who. Finance Systems implements; AP Manager requests; Control owner approves.

What can go wrong. Shared human credentials. Production write “just for the demo.” Prompt-time instructions used as the only access control.

Control. Technical enforcement preferred over procedure. If the platform cannot enforce a cap, autonomy stays at L1 and the residual risk is logged.

Measure. Entitlements vs charter (drift count); unused write verbs.

Evidence. Identity sheet attached to the charter.

4. Approval boundaries

What to do. Agents do not create DOA. They may route, remind, or prepare. Approval remains a human act unless a **separate**, documented workflow engine already performs deterministic routing — and even then the agent does not become the approver.

How.

- DOA table is a source of truth owned by the Controller.
- Agent may flag `EX-APR-001` / `EX-APR-002` .
- Agent may not approve, re-assign DOA, or raise a limit.
- Value caps on agent **actions** (`send`, `park`) are not a substitute for DOA on the invoice.

Who. Controller owns DOA. Approval Agent’s Accountable is Head of AP.

What can go wrong. Auto-approval “because the match is clean” introduced in a prompt. Delegation harvested from out-of-office text and treated as DOA.

Control. Prompt and workflow deny-lists for approve/release verbs. Sample of agent-touched approvals in QA.

Measure. Agent-initiated approval events (target: zero unless a documented non-agent workflow is in scope — even then, count and review).

Evidence. DOA version cited on the charter; QA sample.

5. Role-based access (RBAC)

What to do. Map people and agents to named roles. No personal standing exceptions without expiry.

How. Human roles follow `04_RACI_STANDARD.md`. Agent roles are `agt.[charter].[env]` (e.g. `agt.match.shadow`). Privileged break-glass is a separate role, logged, time-bound (`03_INCIDENT_AND_OVERRIDE.md`).

Who. Finance Systems + Control owner.

What can go wrong. Copy-paste of a privileged role to “make the pilot work.”

Control. Joiner/mover/leaver for agent identities as well as people. Quarterly RBAC certification.

Measure. Standing exceptions; stale identities.

Evidence. RBAC matrix; certification.

6. Data privacy

What to do. Treat invoice, vendor, employee and bank data as personal or confidential by default. Minimise what enters prompts, logs, vendor clouds and training sets.

How.

- Data inventory: fields the agent may read, may write to evidence, may not send to a model vendor.
- Redaction rules from `../Process_Mapping/02_TRANSCRIPT_TO_WORKFLOW.md` apply to prompts and traces.
- Buyer records lawful basis and vendor location in `[BUYER privacy file]`. This OS does not declare GDPR/CCPA adequacy.
- Default: **no** customer data used to train a shared vendor model.

Who. Privacy / DPO Accountable for the inventory. AP Manager Responsible for not pasting restricted data into unsanctioned tools.

What can go wrong. Full invoice PDF pasted into a consumer chatbot. Traces stored with IBAN in plain text.

Control. Approved tool list. Technical DLP where the buyer has it. Privacy review of each charter.

Measure. Unsanctioned-tool incidents; traces found with forbidden fields.

Evidence. Inventory; vendor processing record; privacy review on the charter.

7. Confidential information

What to do. Separate ordinary AP data from marked confidential (M&A vendors, sealed settlements, employee-payable, attorney-client).

How. Confidential suppliers/entities on a deny-list or a dedicated human-only queue. Agents default to **exclude** until an allow-list is signed.

Who. Legal + Controller define the list. AP Manager operates the queue.

What can go wrong. Orchestrator prioritises a sealed invoice into a shared agent Slack.

Control. Deny-list enforced in routing. Periodic search of evidence stores for listed vendor IDs.

Measure. Confidential items processed by a general agent (incident).

Evidence. Deny-list version; routing logs.

8. Prompt injection

What to do. Treat every invoice, email, statement, PDF comment, and filename as **untrusted text**.

How.

- Untrusted content is data, never instructions. System prompt states that documents cannot change rules, autonomy, or recipients.
- Outbound actions (mail, post, master-data write) require a separate policy check that does not read the document body for instructions.
- Hidden text / very small text / "ignore previous" patterns are flagged `EX-QLT-001` or `EX-SYS-001` for human review.
- Tools the agent can call are an allow-list.
- Do not retrieve arbitrary URLs printed on invoices.

Who. Finance Systems + Control owner design; AP Manager escalates suspected injection.

What can go wrong. Invoice line "please pay to new IBAN" processed as a bank change. Email from a supplier that says "approve this exception" consumed as a workflow command.

Control. Injection test cases in every release (`02_RELEASE_AND_CHANGE.md`). Bank-change path is never document-driven (`EX-PAY-001`).

Measure. Injection-test fails; incidents tagged `INJ` .

Evidence. Test file; incident records.

9. Hallucination

What to do. Separate **extracted or computed facts** from **generated language**. Facts that are not in the source or the system of record are defects.

How.

- Deterministic work (match maths, exact duplicate keys, ageing) must not be performed by a language model.
- Model outputs that assert a PO number, amount, VAT ID, or approver must cite the field source. Uncited assertions fail validation.
- Draft communications may not invent discounts, legal positions, or payment dates.
- Root-cause clusters are hypotheses until a human accepts them.

Who. AP Manager owns output standards. Control owner samples.

What can go wrong. Agent invents a GR number. Report shows a saving that Finance did not validate.

Control. Output validation (section 10). QA sample. Dictionary bans unvalidated financial lines.

Measure. Hallucination defects in QA; FNR/FPR per `01_KPI_DICTIONARY.md`.

Evidence. QA file; failed-output store.

10. Output validation

What to do. No agent output becomes an operational action until it passes the validators for that charter.

How. Layer validators:

1. **Schema** — required fields present, types valid, taxonomy code exists.
2. **Cite** — amounts, IDs, dates match source or ERP read.
3. **Policy** — autonomy level, value cap, allow-list, SoD.
4. **Human** — required at L0–L2 for external action; at L3+ for exceptions listed on the charter.

Failed validation = no send/post + exception to human + logged reason.

Who. Finance Systems implements validators. AP Manager handles fails.

What can go wrong. Validator bypass flag left on after debug. Only schema checked.

Control. Bypass is break-glass (`03_INCIDENT_AND_OVERRIDE.md`). Release tests include failed-validation cases.

Measure. Validation-fail rate; bypass events.

Evidence. Validator logs; bypass tickets.

11. Audit logs

What to do. Keep an immutable-enough trail of what the agent saw (reference, not necessarily the full payload), what it proposed, what was validated, who accepted, and what system action occurred.

How. Minimum event fields: timestamp (UTC), agent identity, charter version, model/prompt hash, input references, output hash, validation result, human actor, action verb, correlation ID, invoice/vendor IDs. Retention in section 21. Logs are not writable by the agent identity.

Who. Finance Systems operates. Control owner and Internal Audit read.

What can go wrong. Logs only on the vendor side, deleted at contract end. Logging the full IBAN "for completeness."

Control. Log-integrity check. Privacy review of log fields.

Measure. Events without correlation ID; gaps vs invoice volume.

Evidence. Log store pointer; integrity report.

12. Version control

What to do. Version every artefact that can change behaviour: SOP, RACI, charter, prompt, tools list, taxonomy, validators, DOA citation.

How. Semantic versions. Production runtime cites hashes (prompt, model ID, workflow). "Latest" is not a version.

Who. AP Manager (content); Finance Systems (runtime hashes).

What can go wrong. Prompt edited in the vendor UI with no record.

Control. Production run refuses to start without hash match to the approved release (02_RELEASE_AND_CHANGE.md).

Measure. Unmatched runtime hashes (incident).

Evidence. Version register.

13. Model changes

What to do. Treat a model swap, major version, or temperature/tool-config change as a **release**, not a maintenance patch.

How. See 02_RELEASE_AND_CHANGE.md . Re-run the labelled Test pack. Shadow if error profile moves. No model change combined with an autonomy promotion.

Who. Finance Systems proposes; Control owner + Head of AP approve.

What can go wrong. Vendor silently routes to a new model. "It's the same family."

Control. Pin model IDs. Alert on vendor-side ID change.

Measure. Unplanned model-ID changes; post-change defect delta.

Evidence. Release record; pin configuration.

14. Workflow changes

What to do. Treat routing, allow-lists, value caps, exception-code mapping and validator rules as workflow changes — same discipline as model changes.

How. Change ticket; impact on SoD and DOA; test cases for the touched codes; communication to processors.

Who. AP Manager requests; Control owner approves policy-affecting changes; Finance Systems implements.

What can go wrong. Someone widens tolerance in a config table to "clear the queue."

Control. Config write restricted. Diff of production workflow vs last approved export, periodic.

Measure. Unapproved config diffs.

Evidence. Export + ticket.

15. Testing

What to do. No production action rights without a labelled historical pack and a recorded go/no-go (`00_METHODOLOGY.md` Step 6).

How. Labels before model output. Stratify by taxonomy code and high-risk types (duplicates, bank change, tax, high value). Include injection and hallucination cases. Buyer sets numeric gates; this file does not.

Who. Analyst + AP specialists label; Control owner accepts the scoring rules.

What can go wrong. Vendor demo set used as the only test.

Control. Test-pack ownership is the buyer's. Promotion checklist requires the pack ID.

Measure. Codes with $n = 0$ in the pack; gate breaches.

Evidence. `[BUYER]/Evidence/Test/`.

16. Release management

What to do. Ship agent behaviour through the same change path as any finance system change.

How. `02_RELEASE_AND_CHANGE.md` : change record, back-out, owners, windows, production hash lock.

Who. Finance Systems + Head of AP. Payment-adjacent releases Inform Treasury.

What can go wrong. Weekend “hot fix” to a prompt.

Control. Emergency changes still ticketed, with 24-hour retrospective.

Measure. Emergency vs planned releases; rolled-back releases.

Evidence. Change records.

17. Incident response

What to do. When an agent causes, or could have caused, a wrong payment, data leak, control bypass, or operational outage, follow `03_INCIDENT_AND_OVERRIDE.md`.

How. Detect, contain (disable identity / revoke send), eradicate (fix + test), recover, review. Classify severity. Notify Accountable within the buyer’s clock.

Who. AP Manager coordinates; Control owner classifies; Finance Systems contains technical identities; Controller involved when money or books are affected.

What can go wrong. Incident closed as “user training” with the agent still live.

Control. Severity table; time-to-contain metric; no close without residual-risk statement.

Measure. Incidents by severity; open > buyer SLA; repeat incidents.

Evidence. Incident file.

18. Override

What to do. Allow a named human to bypass an agent recommendation or validator **with a reason code**, never silently.

How. Override types: accept agent (agree), reject agent (proceed differently), force-path (bypass validator — break-glass). Force-path requires Control owner or Head of AP per severity. All overrides logged and sampled.

Who. Processors may reject recommendations. Only listed roles force-path.

What can go wrong. Override becomes the operating model.

Control. Override rate on the weekly report. Force-path dual control.

Measure. Override rate; force-path count; codes most overridden (signal of a bad rule, not a bad user).

Evidence. Override log.

19. Fallback

- What to do.** Pre-write how AP continues when the agent, model vendor, or interface is down.
- How.** Fallback is the CS-1.0 SOP without the agent lane. Queues must remain workable in the ERP/workflow the humans already use. Do not make the agent the system of record.
- Who.** AP Manager owns fallback drills. Finance Systems owns vendor-down detection.
- What can go wrong.** Work only exists in the agent UI.
- Control.** Semiannual fallback drill. Agent UIs are not the archive.
- Measure.** Drill success; minutes to disable an agent identity.
- Evidence.** Drill record; disable runbook.

20. Business continuity (BCP)

- What to do.** Include agent vendors and identity stores in the AP BCP. Recovery objective is buyer-set.
- How.** Dependencies: model API, capture, ERP, evidence store, IdP. If the evidence store is down, new agent actions stop (cannot meet audit-log rule). Payment authorisation path must work without agents.
- Who.** Controller / BCP owner Accountable. Finance Systems Responsible for technical dependencies.
- What can go wrong.** Ransomware on the evidence store with no export. Vendor bankruptcy.
- Control.** Dependency map; export of evidence; exit clause notes in vendor file (Legal).
- Measure.** Dependencies without an owner; last successful evidence export.
- Evidence.** BCP annex; export logs.

21. Evidence retention

- What to do.** Keep proof for the period the buyer’s finance, tax and litigation policies require. This OS does not set a statutory period.
- How.** Retention table (buyer completes periods):

Class	Examples	Period BUYER	Store
Operating evidence	Charters, logs, QA, scorecards		
Tax-relevant invoices / workpapers	As finance policy		

Class	Examples	Period [BUYER]	Store
Privacy-limited traces	Prompts with residual PII	Minimise; not longer than operating need	Restricted
Incident / override			

Disposal is logged. Agents cannot delete evidence.

Who. Controller owns retention policy. Privacy owns personal-data minimisation.

What can go wrong. Infinite vendor logs; or 30-day wipe that destroys audit proof.

Control. Annual retention attestation.

Measure. Stores past policy; stores with no policy row.

Evidence. Retention schedule; attestation.

22. Access termination

What to do. Remove human and agent access the same day the need ends: leaver, role change, charter retired, vendor off-boarded, pilot ended.

How. Checklist: IdP, ERP roles, workflow, mailbox, vendor console, API keys, evidence-store ACLs, shared vaults. Confirm no standing personal tokens.

Who. Line manager + Finance Systems. Control owner samples.

What can go wrong. Agent key in a personal vault after the vendor project ends.

Control. Leaver ticket cannot close without agent-identity section. Quarterly orphan-identity scan.

Measure. Orphan identities; tickets closed without checklist.

Evidence. Termination tickets; scan report.

23. Periodic certification

What to do. Recertify, on a calendar, that the operating facts are still true.

Item	Default frequency (buyer may tighten)	Certifier
Incumbents and deputies	Quarterly	Head of AP
RBAC + agent entitlements	Quarterly	Control owner + FinSys
SoD exceptions	Quarterly or at expiry	Controller
Charters vs runtime hashes	Monthly	FinSys + AP Manager
Confidential deny-list	Quarterly	Legal + Controller

Item	Default frequency (buyer may tighten)	Certifier
Retention / privacy inventory	Annual	Privacy + Controller
Fallback drill	Semiannual	AP Manager
Control matrix still mapped	Semiannual	Control owner

How. Attest “accurate / exception listed.” Exceptions enter the risk register.

Who. As table. Internal Audit may inspect the file; they do not own the operating cert.

What can go wrong. Click-through certification.

Control. Sample of entitlements vs ERP extract on each cycle.

Measure. Overdue certs; exception rate.

Evidence. Certification pack.

24. Vendor and model risk

What to do. Treat model hosts, OCR vendors, iPaaS and “agent platforms” as third parties that can change behaviour, fail, or retain data.

How. Minimum vendor file: services in use, data classes sent, residency, subprocessors, model-pin capability, silent-update policy, incident contact, exit/export, whether buyer data is used for training (must be **no** unless Controller + Privacy sign). Concentration risk: a single vendor that is both capture and payment messaging is recorded.

This is **not** a vendor-assurance certificate and not a security audit.

Who. Finance Systems + Procurement / vendor management. Privacy and Legal on data terms. Control owner on operational residual risk.

What can go wrong. Shadow SaaS. Training-data opt-in buried in a UI. Model change with no notice.

Control. Approved-vendor list. Charter cannot cite an unlisted tool. Pin + monitor model IDs.

Measure. Unlisted tools found; vendors without an exit/export test.

Evidence. Vendor file; list; last export test.

Governance operating cycle

Cadence	Forum	Inputs	Decisions
Daily	AP Manager huddle	Queue, incidents, overrides	Containment
Weekly		Scorecard	Hold / investigate

Cadence	Forum	Inputs	Decisions
	Agent performance (03_WEEKLY_AGENT_REPORT.md)		
Monthly	Hash and entitlement check	Runtime vs charter	Emergency disable if drift
Quarterly	Certification	RBAC, SoD, incumbents	Exceptions and expiries
Each release	Change record	Test pack	Go / no-go
Each promotion	Autonomy board (can be the same people as change)	Measure pack	One increment or refuse

What this framework is not

- A legal opinion, SOC report, or ISO certificate.
- A guarantee that fraud will be detected or that books are correct.
- Permission to process payments without the buyer’s existing authorisation path.

Proof before permission.

End of 00_GOVERNANCE_FRAMEWORK.md

Evidence Room — AP Agent OS Professional

Controls — 00 Control Framework

Product: Evidence Room AP Agent OS — Professional
Module: Controls
Standard: Proof before permission
Audience: Control / Risk owner, Head of AP, Controller, Internal Audit, Finance Systems
ERP stance: Agnostic. Controls attach to duties and evidence, not to a named module.
Examples: ACME Industrial Holdings — **ILLUSTRATIVE** only.
Version: 1.0
Classification: Design standard for AP agent controls. Does not certify an audit opinion or regulatory compliance.

Purpose

Design, operate and evidence controls around the 16-agent stack so that errors are prevented or detected **before** they become payments, master-data changes, or unsubstantiated reports.

This framework sits on `../Governance/00_GOVERNANCE_FRAMEWORK.md` . It does not replace ERP inherent controls. Agents must not be used to bypass them.

1. What to do

For every material risk in `02_RISK_REGISTER.md` and every agent in `01_CONTROL_MATRIX.md` :

- 1. Name the risk in operational language.
- 2. Assign a control with a human owner.
- 3. Mark Preventive or Detective (or both, as two rows).
- 4. Define evidence and frequency.
- 5. Define the escalation trigger.
- 6. Map residual risk if the control is procedural-only (not technically enforced).

Do not list “training” as the only control for a money-adjacent risk.

2. How

2.1 Control objectives (AP agent OS)

ID	Objective	Typical failures if absent
CO-ACC	A named human is Accountable for each agent and exception family	“The model paid it”
CO-SOD	Incompatible duties stay incompatible for people and identities	Service account releases payment
CO-PRV	Least privilege; allow-lists; no standing write beyond charter	Scope creep
CO-VAL	Outputs are schema-, cite-, and policy-validated before action	Hallucinated PO / IBAN
CO-INJ	Untrusted document text cannot change rules or recipients	Invoice-as-instruction
CO-DUP	Exact and potential duplicates are handled before payment	Double pay
CO-MAT	Match and tolerance follow documented policy, not a prompt	Silent tolerance widen
CO-MDM	Vendor, bank, entity, tax standing data change is human-controlled	Document-driven bank change
CO-APR	DOA remains human; agents route/remind only	Prompt auto-approve
CO-PAY	Payment authorisation remains human	Proposal edited by agent
CO-LOG	Actions are attributable and retained	No trail
CO-CHG	Behaviour changes are released, not hot-edited	Unhashed prompt
CO-OVR	Overrides are logged and trended	Force-path as process
CO-BCP	AP operates without the agent	Work trapped in a bot UI

ID	Objective	Typical failures if absent
CO-MSR	Promotion uses dictionary KPIs, not activity	Vanity go-live
CO-TAX	Tax and entity treatment is not invented by a model	Wrong legal entity / tax

2.2 Preventive vs detective

Type	Meaning	Agent implication
Preventive	Stops the action	Validators, allow-lists, denied verbs, SoD on the identity
Detective	Finds the error after proposal or after posting	QA sample, duplicate after-the-fact, statement rec, ageing review

Prefer preventive for bank data, payment release, entity, and tax posting. Detective-only for those is a residual risk that must appear on the register.

2.3 Control design pattern (every control)

Use the seven answers:

```

What to do: [the check]
How:       [system rule / sample method]
Who:       [human owner – never the agent]
What can go wrong: [bypass, rubber-stamp, wrong sample]
Control of the control: [who tests this control]
Measure:    [dictionary ID or count]
Evidence:   [artefact]

```

2.4 Mapping to exception codes

Exception taxonomy (03_EXCEPTION_TAXONOMY.md) is a **detection and routing** structure. It is not itself a control. A control exists when someone (or a validator) must do something reliable with that code.

Example: EX-DUP-001 raised by an agent is a detective signal. The control is: payment preparer cannot include a line with open EX-DUP-001 without OV-FCE.

2.5 Sampling

Where frequency is "sample," the buyer records n, method (random / stratified by value / stratified by code), and who draws the sample. Drawing the sample after seeing the agent's "good" items is forbidden.

2.6 Layers (defence in depth)

1. ERP / workflow inherent (DOA, park, 3-way match if configured)
2. Agent identity entitlements
3. Validators (schema / cite / policy)
4. Human R at the autonomy level
5. Payment-proposal review
6. Statement rec and close completeness

7. QA / Internal Audit sample

An agent that “also checks” does not allow removal of layer 1.

3. Who

Role	Duty
Control / Risk owner	Accountable for this framework and the matrix
Head of AP	Responsible for operating the controls day to day
Controller	Money, close, DOA, financial scorecard lines
Finance Systems	Technical enforcement and logs
Exception owners	Controls attached to their codes
Internal Audit	Independent test — not the control owner
Agents	May perform a detective procedure (flag). May not own a control

4. What can go wrong

Failure	Effect
Matrix copied from a vendor brochure	Controls nobody operates
Agent listed as control owner	Governance breach
Only activity KPIs monitored	Harmful STP looks “efficient”
Preventive control is a paragraph in a prompt	Not a control
Duplicate sampling of easy invoices	False assurance
ERP control switched off “because the agent matches”	Control regression

5. Control (of this framework)

- Semiannual matrix recertification (`../Governance/00_GOVERNANCE_FRAMEWORK.md §23`).
- Every matrix row has a human owner and an evidence pointer.
- Every residual “procedural only” control has a risk-register ID.
- Internal Audit may test a sample of rows; findings open issues, they do not silently edit the matrix.

6. Measure

Metric	Definition
Matrix completeness	Agents with ≥ 1 row / 16
Owner vacancies	Rows with no named incumbent
Overdue control operation	Frequency missed
Control breaches	Dictionary: control breaches
Escalations fired vs should have fired	Sample
Audit exceptions	Dictionary

7. Evidence

- This framework (versioned)
- 01_CONTROL_MATRIX.md operating copy with local names
- Control operation artefacts (QA files, access reviews, proposal checklists)
- Certification pack

Store under [BUYER]/Evidence/Controls/ .

How to extend the matrix

When adding a 17th agent or a new verb:

1. Add risks to 02_RISK_REGISTER.md .
2. Add matrix rows (never "covered by existing culture").
3. Map exception codes if new stops-the-line conditions exist.
4. Issue a C2/C3 change.
5. Do not promote autonomy in the same change.

ACME — ILLUSTRATIVE

ACME keeps ERP three-way match on. Matching Agent at L1 explains variances; it cannot tick "match complete." Payment Proposal Review Agent at L1 cannot remove a hold. Control owner refuses a request to disable ERP duplicate check "to reduce false positives."

Related documents

- 01_CONTROL_MATRIX.md
 - 02_RISK_REGISTER.md
 - 03_EXCEPTION_TAXONOMY.md
 - ../Governance/
 - ../KPI_and_Measurement/01_KPI_DICTIONARY.md
-

End of 00_CONTROL_FRAMEWORK.md

Evidence Room — AP Agent OS Professional

Controls — 03 Exception Taxonomy

Product: Evidence Room AP Agent OS — Professional

Module: Controls

Standard: Proof before permission

Audience: AP Manager, exception owners, processors, Control owner, agent designers

ERP stance: Agnostic. Map each code to the buyer's reason-code field. Do not run a parallel unofficial list.

Examples: ACME notes — **ILLUSTRATIVE**.

Version: 1.0

Purpose

One language for why an invoice is not straight-through. Every parked item, chase, hold and report line uses a code from this file.

Rules of use

1. **Primary code** = what stops posting or payment now. Secondary codes are allowed as tags, not as a dump into "Other."
2. **Do not invent codes** in a queue. If needed, raise a C2 change to this file.
3. **UNMAPPED** is a holding status with a Q-row owner, not a permanent code.
4. Risk **C** = Critical (money-direction, bank, identity abuse). **H/M/L** = High / Medium / Low operational impact. Buyer may re-rate; record the reason.
5. Automation potential is about **resolution**, not detection. Detection may be High while resolution stays Low.
6. Potential agent is a **candidate**, not permission to act. Autonomy stays with the charter.
7. This taxonomy is not a fraud typology and not a legal classification.

Each code uses the same card: definition, probable root cause, required data, suggested resolution, responsible party, escalation, potential agent, automation potential, risk level — plus what can go wrong / control / measure / evidence.

Index

Code	Name	Risk	Automation (resolution)
EX-PO-001	Missing PO	M	Med
EX-PO-002	Invalid PO	M	Med
EX-PO-003	PO closed	M	Low-Med
EX-PO-004	PO exhausted	M	Med
EX-MAT-001	Price mismatch	M	Med
EX-MAT-002	Quantity mismatch	M	Med
EX-GR-001	Missing receipt	M	Med
EX-GR-002	Partial receipt	L-M	Med
EX-DUP-001	Duplicate invoice	H	High (detect) / Low (void)
EX-DUP-002	Potential duplicate	H	Med
EX-MDM-001	Wrong supplier	H	Low
EX-MDM-002	Incorrect legal entity	H	Low
EX-TAX-001	Tax issue	H	Low
EX-APR-001	Approval missing	M	Med
EX-APR-002	DOA issue	H	Low
EX-COD-001	Coding missing	M	Med
EX-COD-002	Invalid cost centre	M	Med
EX-QLT-001	Invoice quality	M	Med
EX-QLT-002	OCR / extraction issue	M	Med
EX-MDM-003	Master-data issue	M	Low
EX-PAY-001	Banking-change concern	C	Low
EX-CN-001	Credit note required	M	Med
EX-STM-001	Statement discrepancy	M	Med
EX-PAY-002	Payment hold	H	Low
EX-DSP-001	Disputed invoice	H	Low
EX-AGE-001	Aged unresolved item	M	Med

Code	Name	Risk	Automation (resolution)
EX-SYS-001	System / interface error	H	Low

EX-PO-001 — Missing PO

Field	Content
Code	EX-PO-001
Definition	Invoice is in a PO-required category (per buyer policy) and no usable PO reference is present, or the reference is blank after validation.
Probable root cause	Non-PO spend; supplier omitted the PO; requester bypassed procurement; channel lost the reference.
Required data	Invoice; supplier; spend category / account hint; requester if known; policy: PO-required flag; value.
Suggested resolution	Obtain a valid PO or follow the documented non-PO path (do not invent one). Do not manufacture a PO number.
Responsible party	Requester / Buyer R ; Exception owner coordinates; Procurement A for after-the-fact PO policy.
Escalation	Repeat supplier or requester; value $\geq$ [BUYER cap] ; aged past [BUYER clock] .
Potential agent	Exception Triage (classify); Supplier Resolution (draft request for PO#); Internal Follow-Up (requester); PO Quality (repeat bypass).
Automation potential	Med — request drafts High; creating a PO Low .
Risk level	M (H if high value or restricted category).

What can go wrong. After-the-fact PO used to hide bypass. **Control.** Retro-PO flagged, not auto-cleared. **Measure.** Volume, repeat rate, age. **Evidence.** Exception record + policy path taken.

ACME — ILLUSTRATIVE. Marketing ads invoice, no PO. Routed to non-PO workflow; not parked as “wait for any PO.”

EX-PO-002 — Invalid PO

Field	Content
Code	EX-PO-002
Definition	A PO number is present but is not valid for this invoice: unknown, other supplier, other entity, cancelled, or wrong document type.
Probable root cause	Typo; supplier reused an old PO; cross-entity quote; cancelled PO still printed on the template.

Field	Content
Required data	Invoice PO string; ERP PO header (supplier, entity, status); supplier ID.
Suggested resolution	Correct the reference if a valid PO exists; else treat as EX-PO-001 / policy path. Do not switch supplier on the PO.
Responsible party	Buyer R ; AP Exception owner coordinates.
Escalation	Entity mismatch (also tag EX-MDM-002); supplier mismatch (tag EX-MDM-001).
Potential agent	Matching (deterministic lookup); Exception Triage; Supplier Resolution (ask for correct PO#).
Automation potential	Med
Risk level	M

What can go wrong. Agent “finds a PO that fits” by amount. **Control.** PO must match supplier + entity, not only amount. **Measure.** Invalid-PO rate by supplier. **Evidence.** Lookup log.

EX-PO-003 — PO closed

Field	Content
Code	EX-PO-003
Definition	Cited PO exists but is closed / finally invoiced / blocked for invoicing per ERP status.
Probable root cause	Late invoice; duplicate shipment invoicing; premature close; residual service on a closed order.
Required data	PO status, close date, close reason, invoiced-to-date, invoice date, buyer.
Suggested resolution	Buyer decides: reopen per policy, new PO, or reject / credit. AP does not reopen silently.
Responsible party	Buyer R ; Procurement policy A .
Escalation	Value ≥ cap; supplier dispute (EX-DSP-001).
Potential agent	Matching; PO Quality; Internal Follow-Up; Supplier Resolution if reject/credit.
Automation potential	Low–Med (status detect High; reopen Low).
Risk level	M

What can go wrong. Reopen used as a habit. **Control.** Reopen is a logged Procurement action. **Measure.** Reopen count vs rejects. **Evidence.** Status snapshot at exception time.

EX-PO-004 — PO exhausted

Field	Content
Code	EX-PO-004
Definition	PO is open but remaining quantity or value is insufficient for this invoice (after documented tolerance).
Probable root cause	Under-ordered; price increase; extra lines; prior invoices consumed the PO; blanket PO poorly sized.
Required data	PO remaining qty/value; invoice qty/value; prior invoices; tolerance policy version.
Suggested resolution	Change-order / new PO / split per policy; or credit. Do not raise tolerance to fit.
Responsible party	Buyer R .
Escalation	Repeat exhaust on same PO; blanket PO pattern (PO Quality).
Potential agent	Matching (deterministic remaining); PO Quality; Internal Follow-Up.
Automation potential	Med
Risk level	M

What can go wrong. Tolerance widened in config. **Control.** Config lock (01_CONTROL_MATRIX.md).
Measure. Exhaust rate by buyer/supplier. **Evidence.** Remaining-balance snapshot.

EX-MAT-001 — Price mismatch

Field	Content
Code	EX-MAT-001
Definition	Invoice price (line or header, per policy) differs from PO/contract price beyond the documented tolerance.
Probable root cause	Price change not on PO; tax/freight in unit price; currency; UoM; supplier error; outdated PO.
Required data	Invoice price; PO/contract price; currency; UoM; tolerance table; freight/tax treatment rule.
Suggested resolution	Buyer accepts with documented authority, or supplier revises / credits (EX-CN-001). Deterministic compare only.
Responsible party	Buyer R ; Exception owner coordinates.
Escalation	Variance $\geq$ [BUYER] value or %; restricted items.
Potential agent	Matching (compare + explain); Exception Triage; Supplier Resolution; PO Quality if PO stale.

Field	Content
Automation potential	Med
Risk level	M (H if systematic overcharge).

What can go wrong. Model “considers it close.” **Control.** Arithmetic compare; model cannot pass. **Measure.** Variance distribution (descriptive). **Evidence.** Compare log.

EX-MAT-002 — Quantity mismatch

Field	Content
Code	EX-MAT-002
Definition	Invoice quantity differs from PO and/or receipted quantity beyond documented tolerance (two- or three-way per policy).
Probable root cause	Partial delivery invoiced in full; UoM; over-shipment; service hours vs order; duplicate shipment.
Required data	Invoice qty; PO qty; GR qty; UoM; prior invoices; tolerance.
Suggested resolution	Wait/align GR (see EX-GR-*); supplier credit; PO change. Do not receipt to match unless GR owner independently confirms receipt.
Responsible party	GR owner and/or Buyer depending on whether goods arrived.
Escalation	Over-invoice qty with no GR; repeat supplier.
Potential agent	Matching; Goods Receipt; Supplier Resolution.
Automation potential	Med
Risk level	M

What can go wrong. Fake GR. **Control.** Agent cannot create GR. **Measure.** Qty-mismatch vs GR-create audit. **Evidence.** Qty snapshot.

EX-GR-001 — Missing receipt

Field	Content
Code	EX-GR-001
Definition	Policy requires a goods/service receipt and none exists against the PO/line for this invoice.

Field	Content
Probable root cause	Goods not in; GR not entered; service confirmation lag; wrong PO so GR sits elsewhere; interface lag (consider EX-SYS-001).
Required data	PO; GR records; requester / receiver; delivery note if any; invoice date; interface heartbeat.
Suggested resolution	Confirm physical/service receipt; GR owner receipts; or reject/credit if not received. Chase per SOP clock.
Responsible party	Goods-receipt owner R ; Exception owner chases.
Escalation	Age > [BUYER] ; value ≥ cap; payment week.
Potential agent	Goods Receipt; Internal Follow-Up; Exception Triage; Matching (detect).
Automation potential	Med (chase drafts); GR post Low .
Risk level	M

What can go wrong. Paid with no receipt; or GR raised to clear AP. **Control.** GR-create deny on agents; proposal check for open EX-GR-001 if policy forbids pay-without-GR. **Measure.** Open count, age, missing-receipt reduction (dictionary). **Evidence.** Chase artefacts.

EX-GR-002 — Partial receipt

Field	Content
Code	EX-GR-002
Definition	Receipts exist but are insufficient versus invoice quantity (beyond tolerance), or invoice is for a later delivery not yet receipted.
Probable root cause	Partial delivery; invoice ahead of remaining goods; multiple GR expected.
Required data	Cumulative GR vs invoice vs PO; delivery schedule if any.
Suggested resolution	Park pending remaining GR; or part-process only if ERP and policy allow; else credit for over-invoiced portion.
Responsible party	GR owner R .
Escalation	Remaining GR silent past clock; supplier invoices 100% repeatedly.
Potential agent	Matching; Goods Receipt; Supplier Resolution.
Automation potential	Med
Risk level	L–M

What can go wrong. Full pay on partial receive. **Control.** Match rule uses cumulative GR. **Measure.** Partial-receipt age. **Evidence.** Cumulative GR snapshot.

EX-DUP-001 — Duplicate invoice

Field	Content
Code	EX-DUP-001
Definition	Exact-key duplicate per buyer rule (typical keys: supplier + invoice number + entity, optionally amount/date). Already exists posted, parked, or paid.
Probable root cause	Supplier resend; dual channel; credit-and-rebill confusion; clerk re-key.
Required data	Both documents; keys; status (parked/posted/paid); payment reference if any.
Suggested resolution	Stop the later item. Do not pay twice. Void/reject per SOP. If the first was wrong, human documents why the second is the valid one.
Responsible party	Exception owner R ; Payment preparer must not include without OV-FCE.
Escalation	Any paid collision = incident path; high value.
Potential agent	Duplicate & Anomaly; Invoice Validation; Payment Proposal Review (block).
Automation potential	High detect; Low void/pay decision.
Risk level	H

What can go wrong. “Same number, different amount” mishandled — that may be EX-DUP-002.
Control. Proposal block. **Measure.** Duplicates detected; duplicate payments prevented **where measurable.** **Evidence.** Key match log.

EX-DUP-002 — Potential duplicate

Field	Content
Code	EX-DUP-002
Definition	Near-duplicate: similar amount/date/reference/document hash, or invoice-number variation, not meeting exact-key. Indicator, not a finding of wrongdoing.
Probable root cause	Number format change; currency; split invoices; genuine similar billing; possible duplicate.
Required data	Candidate pair; similarity reasons; both images/extracts; payment status.
Suggested resolution	Human pair review. Confirm distinct (release code) or upgrade to EX-DUP-001. Do not allege fraud in supplier mail.
Responsible party	Exception owner R .
Escalation	High value; same pair previously paid; bank-change also present.

Field	Content
Potential agent	Duplicate & Anomaly; Payment Proposal Review.
Automation potential	Med
Risk level	H (because FNR is costly; FPR is noisy).

What can go wrong. FPR floods; staff auto-clear. FNR on number +1 tricks. **Control.** Dual QA of FPR and paid-population FNR sample. **Measure.** FPR, FNR. **Evidence.** Pair file + decision.

EX-MDM-001 — Wrong supplier

Field	Content
Code	EX-MDM-001
Definition	Invoice supplier identity does not match the PO supplier and/or the intended vendor master (name, tax ID, bank owner, entity relationship).
Probable root cause	Lookalike vendor; factoring / assignment; group vs local entity; capture picked the wrong master.
Required data	Invoice legal name, tax ID, address; vendor master; PO supplier; any assignment notice (Legal).
Suggested resolution	Stop. MD steward / Procurement confirm identity. Do not "just pick the vendor that matches the amount." Factoring changes follow Legal + Treasury, not AP convenience.
Responsible party	Master-data steward R ; Buyer if PO supplier is wrong.
Escalation	Bank details also differ → treat with EX-PAY-001. High value.
Potential agent	Invoice Validation (flag); Exception Triage; Payment Proposal Review.
Automation potential	Low
Risk level	H

What can go wrong. Pay the lookalike. **Control.** Identity attributes must match policy set; bank path separate. **Measure.** Incidence; near-miss. **Evidence.** Attribute compare.

EX-MDM-002 — Incorrect legal entity

Field	Content
Code	EX-MDM-002
Definition	

Field	Content
	Bill-to / tax / PO entity is not the entity that should take the invoice under buyer policy.
Probable root cause	Supplier template; shared services capture default; intercompany; wrong mailbox.
Required data	Bill-to; PO entity; tax registration; entity register; channel.
Suggested resolution	Recode / reject for rebill to the correct entity. Do not post to "any group company" to hit STP.
Responsible party	AP Manager coordinates; Controller A for entity policy.
Escalation	Cross-border tax impact; close week.
Potential agent	Invoice Intake (route); Validation; Exception Triage.
Automation potential	Low (detect Med).
Risk level	H

What can go wrong. Wrong VAT return. **Control.** Entity allow-list. **Measure.** Entity-mismatch rate. **Evidence.** Entity snapshot.

EX-TAX-001 — Tax issue

Field	Content
Code	EX-TAX-001
Definition	Tax amount, rate, code, recoverability, or registration on the invoice conflicts with the tax engine / policy, or required tax fields are missing.
Probable root cause	Wrong rate; domestic vs reverse charge; missing VAT ID; extract error; supplier tax setup; entity mismatch.
Required data	Invoice tax block; tax engine result; vendor tax standing; entity tax standing; place-of-supply data as policy requires.
Suggested resolution	Tax owner / designated tax processor disposes. Agent must not override the tax engine. Rebill or credit if the document is wrong.
Responsible party	Tax owner A/R (or designated).
Escalation	Large delta; registration missing; audit query.
Potential agent	Invoice Validation (flag vs engine); Exception Triage.
Automation potential	Low
Risk level	H

What can go wrong. Model “fixes” VAT. **Control.** Engine supremacy. **Measure.** Tax exception rate; override of engine (should be zero by agent). **Evidence.** Engine vs invoice diff.

This code is **not** a tax-compliance certificate.

EX-APR-001 — Approval missing

Field	Content
Code	EX-APR-001
Definition	Required approval is not recorded in the official workflow (non-PO path, residual match, or policy step).
Probable root cause	Approver absent; wrong routing; item stuck; informal email “OK” not in the system.
Required data	Workflow history; DOA table version; approver register; invoice value/category.
Suggested resolution	Route/remind using official DOA. Do not accept chat approval unless policy says how to record it. Agent does not approve.
Responsible party	Named approver R ; AP Manager chases.
Escalation	Stall clock; payment run; approver left the company.
Potential agent	Approval Agent; Internal Follow-Up.
Automation potential	Med (reminders)
Risk level	M

What can go wrong. OOO text used as approval. **Control.** Approve-verb deny; official register only. **Measure.** Stall age; approval cycle. **Evidence.** Workflow timestamps.

EX-APR-002 — DOA issue

Field	Content
Code	EX-APR-002
Definition	An approval exists but is outside Delegation of Authority: insufficient limit, wrong dimension (entity, account, project), self-approval, or expired delegate.
Probable root cause	Stale DOA; split invoices to avoid limit; wrong cost object; self-approve; informal deputy.
Required data	DOA table; approver limits; invoice dimensions; related invoices (split detect if policy requires).
Suggested resolution	Re-approve under current DOA. Investigate splits as policy dictates. Agent cannot raise limits.

Field	Content
Responsible party	Controller A (DOA); current valid approver R .
Escalation	Suspected split; any self-approve; high value.
Potential agent	Approval Agent (flag); Payment Proposal Review; Duplicate & Anomaly (split amounts).
Automation potential	Low
Risk level	H

What can go wrong. Prompt “approve within reason.” **Control.** DOA engine is the source; agent flags only. **Measure.** DOA-fail count. **Evidence.** DOA version + decision.

EX-COD-001 — Coding missing

Field	Content
Code	EX-COD-001
Definition	Required GL / cost object / project / tax code assignment is missing and policy does not allow a default without a human.
Probable root cause	Non-PO without coding; PO missing account assignment; extract failed; requester unknown.
Required data	Invoice; PO account assignment; requester; coding guide.
Suggested resolution	Requester/Buyer codes. Agent may propose from PO or a documented map, never from an undocumented guess.
Responsible party	Requester / Buyer R .
Escalation	Close week; high value; repeat uncoded supplier.
Potential agent	Invoice Validation; Internal Follow-Up; Exception Triage.
Automation potential	Med
Risk level	M

What can go wrong. Dump to a suspense account that never clears. **Control.** Suspense ageing in close. **Measure.** Uncoded age. **Evidence.** Coding source (PO vs human vs map version).

EX-COD-002 — Invalid cost centre

Field	Content
Code	EX-COD-002

Field	Content
Definition	Cost centre (or equivalent cost object) is present but invalid: closed, wrong entity, not allowed for the account, or unknown.
Probable root cause	Stale PO; reorg; extract typo; wrong entity.
Required data	Cost object master (open/closed, entity); invoice/PO coding; reorg map if any.
Suggested resolution	Valid replacement from the budget owner. Do not substitute the “closest open” centre automatically.
Responsible party	Budget owner / requester R ; MD steward if the master is wrong (EX-MDM-003).
Escalation	Entity conflict (EX-MDM-002); close.
Potential agent	Validation (deterministic master check); Internal Follow-Up.
Automation potential	Med (detect High).
Risk level	M

What can go wrong. Auto-map to a default centre that distorts P&L. **Control.** No silent default unless documented. **Measure.** Invalid-object rate. **Evidence.** Master snapshot.

EX-QLT-001 — Invoice quality

Field	Content
Code	EX-QLT-001
Definition	Source document is unusable or non-compliant with the buyer's invoice requirements: unreadable, incomplete legal/tax block, wrong document type, or hostile/hidden text.
Probable root cause	Scan quality; statement sent as invoice; pro-forma; missing legal marks; injection-style hidden text.
Required data	Source file; capture report; requirement checklist; if injection suspected — raw text extract.
Suggested resolution	Reject to supplier for a usable invoice, or quarantine if injection suspected (<code>../Governance/03_INCIDENT_AND_OVERRIDE.md</code>). Do not draw missing legal fields from a previous invoice.
Responsible party	AP Processor / Exception owner; Control owner if injection.
Escalation	Injection indicators; repeat supplier quality; missing tax legals.
Potential agent	Invoice Intake; Supplier Resolution (draft reject); Duplicate & Anomaly if “same as last time” reuse is suspected.
Automation potential	Med

Field	Content
Risk level	M (C if confirmed injection with action).

What can go wrong. Completing a document from history. **Control.** No back-fill of identity/bank/tax from prior invoices without human. **Measure.** Quality reject rate by supplier. **Evidence.** Source file + checklist.

EX-QLT-002 — OCR / extraction issue

Field	Content
Code	EX-QLT-002
Definition	Document is readable but extracted fields fail confidence, cite-check, or schema (wrong totals, broken lines, multi-page miss).
Probable root cause	Model/OCR error; complex tables; multi-currency; handwritten notes.
Required data	Source; extract JSON; confidence; cite failures.
Suggested resolution	Human key / correct; or re-extract. Do not send the supplier a letter quoting uncited amounts.
Responsible party	AP Processor R; FinSys if systematic model defect.
Escalation	Model change window; high FNR on amounts.
Potential agent	Invoice Intake; Validation.
Automation potential	Med
Risk level	M

What can go wrong. High confidence, wrong amount. **Control.** Cite-check; QA of high-confidence sample. **Measure.** Extraction accuracy, FPR/FNR. **Evidence.** Extract vs source.

EX-MDM-003 — Master-data issue

Field	Content
Code	EX-MDM-003
Definition	Vendor or related standing data is missing, blocked, incomplete, or inconsistent (tax ID, payment terms, address, withholding) and the issue is not specifically EX-MDM-001/002 or EX-PAY-001.
Probable root cause	Incomplete vendor create; block for compliance/payment; duplicate vendor masters; stale terms.

Field	Content
Required data	Vendor master snapshot; block reasons; request ticket.
Suggested resolution	MD steward amends via official request. AP does not edit master from the invoice. Payment terms changes follow policy.
Responsible party	Master-data steward R .
Escalation	Payment block affecting a critical supplier; duplicate masters with bank differences → EX-PAY-001.
Potential agent	Validation (flag); Internal Follow-Up to steward.
Automation potential	Low
Risk level	M

What can go wrong. Processor edits vendor “just this once.” **Control.** Vendor write denied to processors/agents except steward roles. **Measure.** MDM exception age. **Evidence.** Ticket + before/after (steward).

EX-PAY-001 — Banking-change concern

Field	Content
Code	EX-PAY-001
Definition	Any signal that payment destination may change or is inconsistent: new bank on invoice, callback request, master change in flight, mismatch vs last paid account, or document instructions to change bank. Not a confirmed fraud label.
Probable root cause	Genuine bank change; factoring; extraction error; social-engineering / injection.
Required data	Invoice text; vendor bank master; last payment destination; any change ticket; call-back record per Treasury policy.
Suggested resolution	Hold (EX-PAY-002). Official bank-change path only (Treasury + MD). Never take a new IBAN from the invoice/email body. Human out-of-band verification per buyer policy.
Responsible party	Treasury A ; MD steward R for master; Payment preparer R for hold.
Escalation	Immediate. Treat actioned change from document text as Sev-1.
Potential agent	Payment Proposal Review (flag/hold recommend); Duplicate & Anomaly (unusual payee pattern); Validation (mismatch flag).
Automation potential	Low (detect Med–High).
Risk level	C

What can go wrong. "Helpful" update from PDF. **Control.** Document-driven bank write impossible; hold enforced. **Measure.** Holds placed; incidents. **Evidence.** Hold + official ticket, not the PDF IBAN.

EX-CN-001 — Credit note required

Field	Content
Code	EX-CN-001
Definition	A credit note (or equivalent) is required to correct price, qty, duplicate, tax, or returns — and is not yet on file / matched.
Probable root cause	Agreed return; overbill; cancelled service; tax correction; duplicate already paid.
Required data	Original invoice IDs; reason; expected amount; supplier contact; any debit-balance policy.
Suggested resolution	Request CN (human-released mail); match when received; do not silently net unless policy and ERP support it.
Responsible party	Exception owner R ; Buyer if commercial dispute.
Escalation	Paid overbill; aged CN request; dispute (EX-DSP-001).
Potential agent	Supplier Resolution (draft); Exception Triage; Statement Rec (open CN).
Automation potential	Med
Risk level	M

What can go wrong. Netting without a document. **Control.** CN IDs on the request; human accept send. **Measure.** Open CN requests; age. **Evidence.** Draft + send + CN document.

EX-STM-001 — Statement discrepancy

Field	Content
Code	EX-STM-001
Definition	Supplier statement does not reconcile to AP: missing invoices, missing credits, timing, or unrecognised items.
Probable root cause	Invoice not received; paid but unallocated; timing; wrong account; disputed items.
Required data	Statement; AP open items; payments in period; entity/vendor ID.
Suggested resolution	Work reconciling items: request copies, allocate, or dispute. No GL "plug" from the agent.

Field	Content
Responsible party	AP Manager R .
Escalation	Material unreconciled $\geq$ [BUYER] ; strategic supplier; close.
Potential agent	Vendor Statement Reconciliation; Exception Triage (spawn child codes).
Automation potential	Med
Risk level	M

What can go wrong. Forced balance. **Control.** Adjustments human-only. **Measure.** Unreconciled value/count. **Evidence.** Rec schedule.

EX-PAY-002 — Payment hold

Field	Content
Code	EX-PAY-002
Definition	Item is flagged not to pay: policy hold, bank-change, dispute, missing documents, audit freeze, or manual hold.
Probable root cause	Any high-risk code; supplier query; Treasury instruction; legal.
Required data	Hold reason (must cite a code or official instruction); who placed it; expiry/review date.
Suggested resolution	Remain off proposal until the cited reason clears. Removal requires the role that policy names (not the agent).
Responsible party	Payment preparer operates; reason owner R to clear.
Escalation	Hold removed without reason close; agent removal attempt.
Potential agent	Payment Proposal Review (detect presence); Orchestrator (must not hide holds).
Automation potential	Low
Risk level	H

What can go wrong. Hold lifted to “help cash-out week.” **Control.** Removal logged; agent cannot lift. **Measure.** Holds lifted; reason mix. **Evidence.** Hold audit.

EX-DSP-001 — Disputed invoice

Field	Content
Code	EX-DSP-001

Field	Content
Definition	Business or supplier dispute is open: goods, price, contract, service quality, or legal. Not merely a match fail.
Probable root cause	Commercial disagreement; quality reject; contract interpretation.
Required data	Dispute owner; contract/PO; correspondence; amounts; hold status.
Suggested resolution	Hold payment (EX-PAY-002). Commercial owner leads. AP does not settle via a "goodwill" prompt.
Responsible party	Commercial / Buyer / Legal as policy R ; AP applies hold.
Escalation	Litigation; age; attempted pay.
Potential agent	Exception Triage (tag); Payment Proposal Review (block); Supplier Resolution only with approved language.
Automation potential	Low
Risk level	H

What can go wrong. Orchestrator deprioritises because it is "old." **Control.** Priority floor. **Measure.** Disputed value/age. **Evidence.** Dispute file + hold.

EX-AGE-001 — Aged unresolved item

Field	Content
Code	EX-AGE-001
Definition	Item (invoice, exception, GRNI-related AP item) exceeds the buyer's age threshold and is not in an active controlled path — or the path has stalled. Used in addition to the original code.
Probable root cause	Owner absence; ping-pong; no Accountable; supplier silence; forgotten park.
Required data	Original code; age; last action; owner; value.
Suggested resolution	Re-assign; escalate to the original code's escalation; consider write-off / reject / CN per policy — human.
Responsible party	AP Manager R for the ageing regime; original exception owner still R for content.
Escalation	Crosses next age bucket; close; strategic supplier.
Potential agent	Orchestrator (surface); Root Cause (clusters); Reporting.
Automation potential	Med (surface/chase)
Risk level	M

What can go wrong. Ageing cleared by closing as “no issue.” **Control.** Close above cap is human; age + original code retained. **Measure.** Ageing reduction (dictionary). **Evidence.** Age snapshots.

EX-SYS-001 — System / interface error

Field	Content
Code	EX-SYS-001
Definition	Processing cannot be trusted because a system, job, API, identity, or evidence log failed, is stale, or returned an error. Includes heartbeat failure and suspected injection at the tool layer .
Probable root cause	ERP down; GR interface lag; model vendor 5xx; logger down; clock skew; bad token.
Required data	Job name; time; error; last successful heartbeat; affected population.
Suggested resolution	Fail-closed: do not guess matches on stale GR/PO. Fallback SOP. Incident if action verbs may have fired on bad data.
Responsible party	Finance Systems R ; AP Manager for queue fallback.
Escalation	Payment week; logger down (stop actions); possible wrong-pay.
Potential agent	Orchestrator (surface health); none for “repair.”
Automation potential	Low
Risk level	H

What can go wrong. Agent continues on yesterday’s GR extract. **Control.** Heartbeat; no action if evidence store down. **Measure.** SYS volume; actions without correlation ID. **Evidence.** Job logs.

Cross-cutting operating standard

What to do. Use codes on every non-STP item. Report by code, not by “exceptions.”

How. Map ERP reason codes 1:1. If the ERP allows one reason only, store secondary codes in the Evidence Room.

Who. AP Manager owns the mapping table. Control owner owns this file’s versions.

What can go wrong. “MISC” becomes 40% of the queue.

Control. UNMAPPED and MISC have a weekly cap review. New codes need a C2 change.

Measure. Mix by code; UNMAPPED count; repeat exception rate (dictionary); resolution rate by code.

Evidence. Mapping table version; weekly mix extract.

Mapping table (buyer)

Taxonomy	Local ERP / workflow reason	Owner	Notes
EX-PO-001	[BUYER]		
...			

End of 03_EXCEPTION_TAXONOMY.md

Evidence Room — AP Agent OS Professional

KPI and Measurement — 00 Framework

Product: Evidence Room AP Agent OS — Professional
Module: KPI and Measurement
Standard: Proof before permission
Audience: Head of AP, Controller, AP Manager, Control owner, Finance Systems
Examples: ACME — **ILLUSTRATIVE**. No industry benchmarks are implied.
Version: 1.0
Classification: Measurement standard. Does not promise savings, ROI, or control effectiveness.

Purpose

Measure whether agents are **correct, supervised, cheaper per correct outcome, and safe** — not whether they are busy.
Four families are **reported separately**. They are never netted into a single “AI score.”

Family	Question it answers	May justify promotion?
Activity	What volume did we touch?	No
Operational	Did cycle time, STP, ageing, follow-up clocks improve vs this buyer’s baseline?	Supporting only
Financial	What did inference cost, and what cash or hours were validated ?	Only signed lines
Risk-control	Did we breach, escalate, rework, or miss high-risk items?	Veto — a fail here blocks promotion

1. What to do

1. Freeze a baseline **before** Shadow/Pilot (`../Process_Mapping/00_METHODOLOGY.md` Step 9).
2. Use definitions in `01_KPI_DICTIONARY.md` only.
3. Build weekly agent reports and a management dashboard from those IDs.
4. Set targets with the method in §5 — not from vendor slides.
5. Record gaming risks and the counter-check for every target.

2. How — families

2.1 Activity (descriptive)

Counts of work. Useful for capacity and cost allocation. **Forbidden** as the headline of a steering pack.

Examples: invoices handled, exceptions handled, drafts produced, recommendations issued.

What can go wrong. STP gamed by touching more easy invoices; agents farm volume.

Control. Activity appears below outcomes on every pack. Promotion checklist ignores activity.

2.2 Operational outcomes

Changes in how AP **works**, vs baseline, same scope (entity, channel, type).

Examples: STP rate, exception resolution rate, human intervention rate, average resolution time, time to invoice posting, on-time follow-up, ageing reduction, payment-on-time, missing-receipt reduction, PO compliance improvement, matching/extraction/classification accuracy, FPR, FNR, repeat exception rate.

What can go wrong. Scope mix shift (more easy PO invoices) looks like improvement.

Control. Report mix (channel, type, code). Stratify. Small-n codes show **count**, not a percentage theatre.

2.3 Financial outcomes

Money and cost, in two drawers:

Drawer	Contents	Sign-off
Cost	Cost per invoice, cost per exception resolved, AI inference cost, cost per correct outcome	AP Manager + FinSys (inference ledger)
Validated benefit	Validated financial savings; estimated human hours released (labelled estimate vs validated)	Controller for any cash/P&L line; hours remain labelled

Unvalidated “opportunity” may live in a working appendix marked **NOT FOR STEERING**. It is not a KPI.

What can go wrong. Hours-released × fully-loaded rate published as savings without a headcount or overtime change.

Control. Dictionary fields `validated_by` and `validation_method`. Unsigned financial lines cannot appear on `04_MANAGEMENT_DASHBOARD.md`.

2.4 Risk-control outcomes

Examples: control breaches, escalation rate, audit exceptions, rework rate, override/force-path, hash mismatches, open Sev-1/2, duplicate payments prevented **where measurable**.

A green operational pack with a red risk-control pack is a **failed** period.

What can go wrong. Breaches renamed "user education."

Control. Severity and dictionary IDs; incidents cross-ref.

3. Who

Duty	Accountable	Responsible
Dictionary integrity	Controller	AP Reporting owner / AP Manager
Baseline freeze	Head of AP	Analyst
Weekly report	Head of AP	AP Manager
Financial lines	Controller	AP Manager compiles
Risk-control lines	Control owner	AP Manager compiles
Inference cost ledger	Finance Systems	FinSys
Promotion use of KPIs	Head of AP / CFO as autonomy policy	AP Manager proposes

Agents may draft packs (Reporting Agent L2). They do not accept them.

4. What can go wrong (framework-level)

Failure	Detection
Vendor dashboard replaces dictionary	Pack without dictionary IDs
Baseline never frozen	No baseline worksheet ID on the Measure pack
Mix shift ignored	No mix table
Vanity headline ("1,200 invoices AI-processed")	Reviewer rejects pack
Target copied from another company	No target-method note
Gaming (see dictionary per KPI)	Counter-check blank

5. Target-setting method

Do **not** set a target because a blog cited a percentage.

For each dictionary KPI the buyer chooses to target:

1. **Baseline window** — last complete quarter or [BUYER] periods, same scope.
2. **Minimum n** — below n, no target; report count only.
3. **Direction** — maximise / minimise / guardrail (e.g. FNR has a ceiling).
4. **Guardrails** — a cycle-time target must name the FNR/breach guardrail that cannot worsen.
5. **Method note** — baseline + judgement or capacity model or policy clock. Not industry.
6. **Owner** who can be asked why it moved.
7. **Gaming test** — how we would cheat this target, and the counter-metric.
8. **Review** — targets expire; they are not eternal.

ILLUSTRATIVE ACME method (not a target): ACME sets missing-GR chase on-time as "≥ baseline + improvement the AP Manager will own," with a guardrail that EX-DUP FNR cannot rise. They do **not** set "STP = 80%."

6. Control

- Pack lint: every figure has a dictionary ID, n, scope, period, source.
- Financial drawer locked without Controller.
- Risk-control block mandatory on weekly and management views.
- Measure pack for promotion cites the same IDs (01_AUTONOMY_POLICY.md).
- Changes to formulas are C2 (../Governance/02_RELEASE_AND_CHANGE.md).

7. Measure (of the measurement system)

Check	Definition
ID coverage	Figures on issued packs with a dictionary ID / all figures
Small-n abuse	Percentages published below minimum n
Unsigned money	Count (must be zero on issued packs)
Baseline drift	KPIs whose baseline was edited after freeze without a change ID

8. Evidence

- Baseline worksheet (frozen)
- Dictionary version

- Weekly / management packs
- Inference cost ledger
- Validation sign-offs
- Stored under [BUYER]/Evidence/Measure/

Related documents

- 01_KPI_DICTIONARY.md
 - 02_SCORECARD_GUIDE.md
 - 03_WEEKLY_AGENT_REPORT.md
 - 04_MANAGEMENT_DASHBOARD.md
 - ../Process_Mapping/00_METHODOLOGY.md Step 9
 - ../Governance/01_AUTONOMY_POLICY.md
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End of 00_KPI_FRAMEWORK.md

Evidence Room — AP Agent OS Professional

Business Case — 00 Model

Product: Evidence Room AP Agent OS — Professional

Module: Business Case

Standard: Proof before permission. Evidence decides.

Audience: CFO / Finance Director, Controller, Head of AP, Transformation lead

ERP stance: Agnostic. Costs come from the buyer's ledgers and time study, not from a vendor benchmark.

Examples: ACME Manufacturing — **ILLUSTRATIVE** arithmetic only.

Version: 1.0

Classification: Planning model. **Does not promise savings, ROI, payback, fraud reduction, or compliance.** No line becomes KPI-FIN-SAV until the Controller validates it after operation.

Position

A business case is a **structured estimate** used to decide whether to fund discovery, a first agent, or a wider programme. It is not evidence that an agent works. Evidence is produced in Test → Shadow → Pilot → Measure.

Three scenarios — **Conservative / Base / Upside** — keep you honest. False precision (ROI to two decimals, payback to a week) is a defect.

Use with:

Topic	File
Cost per invoice build	01_COST_PER_INVOICE.md
Inference vs human minutes	02_AGENT_ECONOMICS.md
Phases 0–10	03_IMPLEMENTATION_ROADMAP.md
Dictionary	../KPI_and_Measurement/01_KPI_DICTIONARY.md
Measurement rules	../KPI_and_Measurement/00_KPI_FRAMEWORK.md

Every section: **what / how / who / wrong / control / measure / evidence.**

1. What to do

1. Complete the input sheet from **buyer data** (or mark the input **ASSUMED** and keep it out of the signed case).
2. Compute outputs for Conservative / Base / Upside.
3. Run sensitivity on the three inputs that move the result most.
4. Separate **capacity released** (hours) from **cash** (only where a mechanism exists).
5. Take the case to funding. After go-live, replace estimates with Measure packs.

How. Formulas in §4. Do not paste a blog's "cost per invoice" as your baseline.

Who.

Duty	Accountable	Responsible
Cost model integrity	Controller	FP&A / AP Reporting
Volume and rates	Head of AP	AP Manager
AI / tool / implementation cost	Finance Systems + Procurement	FinSys
Signing the case for funding	CFO / FD or delegated Controller	Transformation lead compiles
Later validation of benefits	Controller	Benefits owner (04_AP_AGENT_OS_TEAM/Implementation/ 02_BENEFITS_REALISATION.md)

What can go wrong. Hours × loaded rate published as P&L savings with no overtime or FTE change. Industry averages invented. Guaranteed ROI language.

Control. Every cash line needs a **mechanism** (overtime down, vacancy not filled, discount actually taken, late fee actually avoided, duplicate actually removed from a proposal). No mechanism → ESTIMATE / capacity only.

Measure. Share of inputs with a source ID; unsigned cash lines (must be zero on the funding pack).

Evidence. Versioned workbook + this narrative under [BUYER]/Evidence/BusinessCase/ .

2. Research context (not Evidence Room results)

Use these **only** as context for why a buyer might invest in measurement, not as default inputs and not as “what AP Agent OS delivers.”

Source	What it reported	How you may use it
Deloitte, 8 Oct 2025, on 2026 outlook	63% of organisations had deployed AI; 21% reported measurable ROI	Label: <i>external research on AI deployment vs measurable ROI — not an Evidence Room outcome.</i> Explains why this OS insists on Measure packs.
The Hackett Group, Nov 2025	Among software adopters studied: 60% touchless processing; 3.5× productivity at ≥30% touchless	Label: <i>software-adopter research, not Evidence Room results, not your baseline.</i> Do not set KPI-OPS-STP = 60% because Hackett published a figure.

Do **not** invent other industry statistics. If you lack a source, leave the cell blank or mark ASSUMED.

3. Inputs (complete every row)

ID	Input	Unit	Source (required)	Conservative	Base	Upside	Buyer value
I-01	Invoice volume	invoices / year	ERP extract				[BUYER]
I-02	AP headcount (in-scope work)	FTE	Org chart / time split				[BUYER]
I-03	Fully loaded labour	currency / FTE / year	FP&A				[BUYER]
I-04	Manual-touch %	% of invoices with a human exception or keying step	Workflow / study				[BUYER]
I-05	Exception rate	exceptions / invoice or % with ≥1 exception	Exception store				[BUYER]
I-06		minutes / exception	Time study or timestamps				[BUYER]

ID	Input	Unit	Source (required)	Conservative	Base	Upside	Buyer value
	Average resolution minutes	(median preferred)					
I-07	Duplicate rate	confirmed EX-DUP-001 / invoices	QA + store				[BUYER]
I-08	Late-payment costs	currency / year actually incurred	AP/Treasury (fees, contractual)				[BUYER]
I-09	Early-pay discount opportunity	currency / year available per terms	Terms extract				[BUYER]
I-10	Processing cost (current)	currency / invoice or / year	01_COST_PER_INVOICE.md				[BUYER]
I-11	AI / tool cost	currency / year (seats, pages, tokens, platform)	Vendor quotes + ledger design				[BUYER]
I-12	Implementation cost	currency (once) + internal days	Roadmap				[BUYER]
I-13	Expected efficiency range	% reduction in in-scope manual minutes	Range , not a point — from Shadow/Pilot evidence if you have it; else ASSUMED and wide	low	mid	high	[BUYER]

Rules for inputs

- I-08: if you cannot see fees actually paid, enter **0** and move discussion to operational ageing — do not estimate “reputation cost.”
- I-09: opportunity is **not** a saving. Saving = discounts **taken** that were not taken before, Controller-validated.
- I-07: do not convert duplicate *flags* into cash. Cash requires proposal-removal or recovered funds (KPI-RSK-DPP / KPI-FIN-SAV).
- I-13: if you have no Shadow/Pilot, Conservative should assume **little or no** minute reduction after adding **new review minutes**.

Who. AP Manager populates volume/rates; FP&A labour; Controller challenges cash-like inputs.

What can go wrong. I-13 set to 50% because a demo was smooth.

Control. ASSUMED cells highlighted. Funding pack lists them on page one.

Measure. Count of ASSUMED vs sourced inputs.

Evidence. Source extract IDs on each row.

4. How — outputs (compute, then label)

All outputs are **model outputs**. They become operational facts only through Measure.

4.1 Baseline cost (annual)

```
Labour_AP = I-02 × I-03
# If I-10 is a fully specified annual processing cost, prefer it over Labour_AP alone.
Baseline_cost = I-10_annual (or Labour_AP + allocated AP systems already in I-10)
```

State what I-10 includes (01_COST_PER_INVOICE.md). Do not double-count.

4.2 Capacity released (hours / year) — ESTIMATE

```
Exception_hours_now = I-01 × I-05 × (I-06 / 60)
# if I-05 is a %, use I-01 × I-05 × I-06 / 60
Manual_hours_now = I-01 × I-04 × (minutes_per_touched_invoice / 60)
# minutes_per_touched_invoice from time study; if missing, do not invent – use
exception_hours only

Hours_released = (Manual_or_exception_hours_now) × I-13
                - new_review_hours
                - new_governance_hours
```

new_review_hours and new_governance_hours are mandatory subtractions. Shadow itself consumes minutes (03_SHADOW_MODE.md).

Label **KPI-FIN-HHR** class: ESTIMATE.

4.3 Processing-cost change (annual, modelled)

```
Processing_cost_after = Baseline_cost
                      - (Hours_released × hourly_loaded)
                      + I-11
                      + (I-12 / amortisation_years)
```

hourly_loaded = I-03 / [BUYER productive hours] (state the hours convention).

Only treat the labour term as cash if a mechanism exists (overtime, contractor, FTE). Otherwise show:

- Line A: capacity (hours)
- Line B: modelled £/\$ if monetised — **NOT FOR STEERING** until Controller agrees the mechanism

4.4 Exception-cost change

```
Exception_cost_now = I-01 × I-05 × (I-06 / 60) × hourly_loaded
Exception_cost_after = same with I-06 and/or I-05 adjusted by I-13
                      + review time on remaining exceptions
```

Pair with quality: cheaper closes that fail QA are not a benefit (KPI-0PS-EXR + rework).

4.5 Estimated savings (three drawers)

Drawer	Contents	Steering?
D1 Capacity	Hours_released	Yes, as hours
D2 Modelled labour \$	Hours × rate if mechanism stated	Only with mechanism
D3 Cash operations	Late fees avoided ($\leq$ I-08, evidenced) + discounts taken ($\leq$ I-09, evidenced) + validated duplicate removals	Only itemised

```
Estimated_savings_model = D2 + D3 - I-11 - amortised I-12
```

Publish D1 always. Publish the model total only with scenario labels.

4.6 Payback (modelled, not a promise)

```
Net_implementation = I-12
Annual_net_benefit_model = D2 + D3 - I-11    (steady year; say so)
Payback_years = Net_implementation / Annual_net_benefit_model
```

If denominator ≤ 0 in Conservative, write **“payback not shown — Conservative net negative or zero.”** Do not invent a payback.

4.7 ROI (modelled, not a promise)

```
ROI_year_1 = (Annual_net_benefit_model - I-12) / I-12
```

Or use a buyer-standard NPV. State the convention. **No false precision:** round money to thousands (or buyer materiality); round ROI to whole percent or “n/m”; round payback to a quarter-year.

4.8 Sensitivity

Vary, one at a time, across a buyer-chosen band:

1. I-13 efficiency
2. I-11 tool cost
3. I-06 resolution minutes (or volume I-01)

Report: which input flips Conservative from negative to positive. That is the management conversation.

5. Scenario rules

Scenario	How to set I-13 and cash drawers	Typical use
Conservative	Low I-13; include full I-11 and I-12; no D3 unless already evidenced; subtract review/governance hours generously	Funding floor; “we still do this for control”
Base	Mid I-13; D3 only with a named mechanism and a haircut (buyer %)	Planning
Upside	High I-13 still inside a range the owner will defend ; D3 still capped by I-08/I-09 actuals	Not the headline

Upside is not a sales case. If Base needs Upside to look acceptable, the project is a control/capacity bet — say that.

Who. Controller refuses a pack that headlines Upside.

Control. Cover page shows Conservative first.

Measure. Presence of all three columns.

Evidence. Scenario tab in the workbook.

6. Worked example — ACME Manufacturing (ILLUSTRATIVE)

Fictional ACME Manufacturing. Figures are **ILLUSTRATIVE arithmetic**, not a market rate and not a result.

Input	ILLUSTRATIVE value
I-01 Volume	8,400 invoices/month → 100,800 / year
I-02 Headcount in scope	9 FTE
I-03 Loaded labour	65,000 / FTE / year
I-04 Manual-touch	70%
I-05 Exception rate	0.45 exceptions / invoice
I-06 Median resolution	18 minutes
I-07 Duplicate confirmed	0.3% of invoices (count only here)
I-08 Late fees actually paid	12,000 / year
I-09 Early-pay opportunity	40,000 / year available on terms
I-10 Processing cost	built in 01 — assume 720,000 / year all-in AP process cost
I-11 AI/tool	48,000 / year
I-12 Implementation (one agent + OS discipline)	90,000 + internal time already in labour

Input	ILLUSTRATIVE value
I-13 Efficiency on exception minutes	Conservative 8% / Base 18% / Upside 30% ASSUMED (no Shadow yet)
Productive hours	1,600 / FTE / year → hourly ≈ 40.6

Exception hours now $\approx 100,800 \times 0.45 \times 18 / 60 \approx 13,608$ hours/year.

Output (rounded)	Conservative	Base	Upside
Hours released after +2,000 review/gov hours	~900	~2,450	~4,080
Monetised labour if overtime/contractor mechanism	~36k	~99k	~166k
D3 cash (Conservative: 0; Base: 25% of I-08 + none of I-09; Upside: 50% I-08 + 25% I-09)	0	3k	16k
Minus I-11	48k	48k	48k
Annual net (before amortising I-12)	negative	~54k	~134k
Payback on 90k	n/m	~1.5–2 years	<1 year
ROI year 1	n/m	do not quote as a promise	do not headline

ACME's Controller would **reject** converting 2,450 hours into 99k savings unless overtime or a vacancy plan exists. The funding conversation is then: **capacity + control evidence**, not ROI.

Hackett's 60% touchless and Deloitte's 21% measurable-ROI figures **do not appear** as ACME inputs.

7. What can go wrong

Failure	Control
Industry average as baseline	Source ID required
Guarantee language	This file forbids it
Duplicate flags as cash	KPI-RSK-DPP rules
Discount opportunity as saving	I-09 vs taken
Ignoring review minutes	Mandatory subtraction
Headline Upside	Cover page order
False precision	Rounding rule
Using the case as Measure	Different folder; dictionary

8. How you measure the case later

Model line	After operation, use
Hours released	KPI-FIN-HHR with time-study ID
Processing cost	KPI-FIN-CPI same model version
Exception cost	KPI-FIN-CPE
Tool cost	KPI-FIN-AIC
Cash	KPI-FIN-SAV only
Quality veto	KPI-ERR-FNR, KPI-RSK-BRH

A project that misses Conservative hours but holds risk-control may still be a **correct** investment. Say so in the Steer pack.

9. Evidence checklist

- ☐ Input sheet with source IDs / ASSUMED flags
- ☐ Three scenarios
- ☐ Sensitivity
- ☐ Mechanism notes for any \$ labour line
- ☐ Research citations labelled as external, unused as inputs
- ☐ Controller review
- ☐ Version and date

Proof before permission.

End of 00_BUSINESS_CASE_MODEL.md

Evidence Room — AP Agent OS Professional

Testing — 00 Testing Standard

Product: Evidence Room AP Agent OS — Professional

Module: Testing

Standard: Proof before permission. Agents earn responsibility. Evidence decides.

Audience: Head of AP, AP Manager, Control owner, Finance Systems, Internal Audit observer, UAT participants

ERP stance: Agnostic. Tests run against the buyer's systems of record and a labelled case file the buyer owns.

Examples: ACME Manufacturing — **ILLUSTRATIVE** only. Not a client, not a result.

Version: 1.0

Classification: Binding test discipline. Buyer sets numeric gates. This file does not certify control effectiveness, legal compliance, or savings.

Position

An agent that has not been tested against labelled historical work is a demonstration, not an operating asset. Testing is how the agent earns the right to be seen by operators (Shadow), then the right to act in a narrow window (Pilot).

Use this standard with:

Topic	File
Historical scripts	01_HISTORICAL_TEST_SCRIPTS.md
UAT forms and scripts	02_UAT_TEMPLATES.md
Live compare, no action	03_SHADOW_MODE.md
Narrow reversible action	04_PILOT_METHODOLOGY.md
Method sequence	../Process_Mapping/00_METHODOLOGY.md Steps 6–8
Change class	../Governance/02_RELEASE_AND_CHANGE.md
Autonomy gates	../Governance/01_AUTONOMY_POLICY.md
Score definitions	../KPI_and_Measurement/01_KPI_DICTIONARY.md
Exception codes	../Controls/03_EXCEPTION_TAXONOMY.md

Every section answers: **what / how / who / what can go wrong / control / measure / evidence.**

1. What testing is for

What to do. Prove four things, in this order, before any production action verb is granted:

1. The agent and the deterministic rules produce the **expected outcome** on cases the buyer already knows.
2. Named operators can **use the output** without inventing a parallel process.
3. On live work, the agent **agrees with humans often enough**, and disagrees in ways the owner can explain.
4. A **narrow, reversible** action set can run without becoming the default path via override.

How. Four stages. Do not skip or merge them into a “soft launch.”

Stage	Question	Action verbs allowed
Historical test	Would it have been right on last quarter's work?	None. Read + score only

Stage	Question	Action verbs allowed
UAT	Can named people operate the designed path?	None in production. UAT environment or paper walkthrough
Shadow	Does it agree with today's humans on live items?	Read + write-evidence only
Controlled pilot	Can a listed verb run inside an allow-list?	Only the listed verbs, caps, entities

Who.

Role	Duty
Head of AP / Controller	Accountable for go/no-go at each stage
AP Manager	Responsible for the test calendar, sample, and daily discipline
Control owner	Accepts scoring rules; vetoes a green pack with a red risk-control line
AP specialists	Create gold labels before seeing agent output
Finance Systems	Provisions non-executing identities; pins hashes
Internal Audit	May observe; does not run the test for the operator

What can go wrong. Vendor demo set used as the only pack. Labels written after the model ran. Production write “just for UAT.”

Control. Promotion checklist in `01_AUTONOMY_POLICY.md` requires a buyer-owned pack ID. Environment confirmation is a gate, not a slide.

Measure. Pack coverage by taxonomy code; labels-before-output attestation; identities with write verbs (must be zero before Pilot).

Evidence. `[BUYER]/Evidence/Test/`, `/UAT/`, `/Shadow/`, `/Pilot/` — each with a signed go/no-go.

2. Operating rules (non-negotiable)

1. **Labels first.** Gold outcomes exist before the agent is run on that item. Relabelling to match the model is a defect.
2. **Buyer owns the pack.** A vendor “golden set” may be *added* as extra cases. It cannot be the only pack.
3. **Stratify.** STP, each material exception family, high value, credit notes, intercompany, FX if used, injection cases, out-of-scope items.
4. **Fail closed.** Timeout, parse fail, missing cite, out-of-scope → no guessed “correct.” Score as fail or `not-comparable`.
5. **Deterministic work is not scored as model brilliance.** Price/qty/tolerance maths must come from the documented rule. If the model “explained” a wrong compare as right, that is a fail.
6. **Buyer sets gates.** Percentages in ACME examples are **ILLUSTRATIVE**. Copying them as targets fails this standard.

7. **Risk-control veto.** A missed EX-DUP-001, EX-PAY-001, or an invented GR is not offset by high STP on easy PO invoices.
8. **One increment.** Test the change you are about to make — not the whole stack.
9. **At-most-once on writes.** When Pilot begins, a timeout is not a retry. Reconcile by read (`../Agent_Library/19_AGENT_CHARTER_STANDARD.md`).
10. **Evidence or it did not happen.** A hallway “it looked good” is not a go.

3. How — environments and identities

What to do. Separate identities for Test, Shadow, and Pilot. Never reuse a human’s ERP login.

How.

Environment	Identity pattern	Allowed verbs	Forbidden
Historical / lab	<code>agt.[charter].test</code>	Read historical extract; write scores to evidence	Any ERP/mail write
UAT	<code>agt.[charter].uat</code>	Read UAT tenant; write UAT evidence	Production data writes
Shadow	<code>agt.[charter].shadow</code>	Read live (or replica); write-evidence	Send, park, post, status change
Pilot	<code>agt.[charter].pilot</code>	Listed verbs only, allow-list enforced	Unlisted verbs; payment release; bank write

If the platform cannot technically block a verb, autonomy stays at L1 and the residual risk is logged on the charter.

Who. Finance Systems implements; Control owner reviews the access extract at the start of each stage and at mid-Shadow.

What can go wrong. Shared “service account” that can park and propose and release. UAT pointed at production.

Control. Access review attached to the stage pack. Mid-Shadow re-extract.

Measure. Count of write entitlements on test/shadow identities (target: zero). Drift vs charter.

Evidence. Access extract + review initials under the stage folder.

4. How — sample design

What to do. Build a labelled file large enough to see the codes you claim to handle, and honest enough to include the codes that hurt you.

How. Minimum design (buyer completes counts):

Stratum	Why it exists	Buyer n (complete)	Notes
Happy-path PO STP	Prevents “we only tested exceptions”	[BUYER]	Include multi-line
EX-PO-*	PO defects	[BUYER]	001-004 if in scope
EX-MAT-*	Price/qty	[BUYER]	Use documented tolerances
EX-GR-*	Receipt	[BUYER]	Partial and missing
EX-DUP-001 / 002	Duplicate	[BUYER]	Include paid population sample
EX-TAX-, EX-MDM-	High-risk standing data	[BUYER]	Small n → report counts, not theatre %
EX-PAY-001	Bank-change bait	[BUYER]	Must appear; expected outcome is human hold
Credit notes / IC / FX	Mix	[BUYER]	If the entity uses them
Out-of-scope / not-an-invoice	Refusal behaviour	[BUYER]	Statements, reminders, marketing
Injection / hostile text	Untrusted content	[BUYER]	See Governance §8
High value ≥ [BUYER cap]	Materiality	[BUYER]	Always human-visible

$n < n_{\min}$ (buyer sets $n_{\min}$): publish **count**, not a percentage.

Who. Analyst pulls IDs; AP specialists label; Control owner confirms high-risk strata are present.

What can go wrong. Last two quiet weeks only. All domestic PO under tolerance.

Control. Coverage matrix signed before the first agent run.

Measure. Codes in charter vs codes with $n = 0$ in the pack.

Evidence. Sample register + coverage matrix in 01_HISTORICAL_TEST_SCRIPTS.md .

5. How — scoring

What to do. Score against dictionary IDs, not vendor dashboards.

How. Required score lines for any agent that classifies or extracts:

Line	Dictionary ID	Rule
Classification accuracy	KPI-ACC-CLS	Taxonomy code or “STP” vs gold
Extraction accuracy	KPI-ACC-EXT	Critical-field list versioned
Matching accuracy	KPI-ACC-MAT	Only if Agent 03 is in the test
False-positive rate	KPI-ERR-FPR	State the condition

Line	Dictionary ID	Rule
False-negative rate	KPI-ERR-FNR	High-risk codes separate; veto eligible
Schema / cite fail	(local count)	Uncited amount/ID = fail
Out-of-scope handling	(local count)	Must refuse or escalate, not invent
Human-required honouring	(local count)	Four human classes never "executed"

Close enough is wrong. Fuzzy invoice numbers, rounded tax that the policy does not allow, and "the vendor is probably this one" are fails.

Disagreements are adjudicated by the **process owner**, not the model vendor.

Who. Analyst scores; process owner adjudicates; Control owner accepts the scorecard.

What can go wrong. Vendor console "98% accuracy" with no field list.

Control. Every reported figure cites a dictionary ID and a pack ID.

Measure. Share of score lines with dictionary IDs (must be 100% on the signed pack).

Evidence. Scorecard + confusion matrices + residual error list.

6. Go / no-go

What to do. Record a binary decision per stage. "Proceed with caution" is a no-go plus a punch-list.

How. The Accountable signs one of:

Decision	Meaning	Next step
Go	Gates met; residual errors listed and accepted	Next stage
Hold	Specific defects; retest those strata	Stay at current stage
Redesign	Charter, rule, or sample is wrong	Return to Agentise or Structure
Stop	Unsafe or out of scope	Disable identity; incident if already live

Buyer writes numeric gates on the charter. This standard only requires that gates **exist**, are **pre-committed**, and are **not edited after seeing results**.

Who. Head of AP / Controller signs Go/Hold/Redesign/Stop. Control owner may force Hold or Stop on risk-control.

What can go wrong. Gates rewritten in the review meeting to keep the project date.

Control. Gate sheet dated and stored **before** the agent run. Diff if anyone edits.

Evidence. Signed decision page in the stage pack.

7. What can go wrong (standard-level)

Failure	What it looks like	Control
Demo-set theatre	40 clean PDFs, no duplicates	Coverage matrix
Label leakage	Specialists saw the model first	Labels-first attestation + timestamp
Production credentials	"UAT" posting live	Identity pattern + access extract
Silent execute	Shadow identity can send mail	Verb allow-list
Mix shift	Easy PO slice called "the test"	Stratification
Offset thinking	Great STP, missed duplicate	Risk-control veto
Retry-all	Pilot double-posts on timeout	At-most-once rule
Date-driven go	Steering wants a demo Friday	Decision enum only

8. How you measure the testing function itself

Question	Metric
Did we test what we claim to run?	Charter codes with n = 0 (should be explained or out of scope)
Did labels precede output?	Attestation + file timestamps
Can we stop?	Minutes to disable the identity (fallback drill)
Did we over-claim?	Scorecard lines without dictionary IDs

9. ACME Manufacturing — ILLUSTRATIVE only

ACME Manufacturing (fictional): one legal entity, email PDF + EDI, domestic PO invoices. Historical pack: 180 labelled items including 12 EX-DUP suspects (3 confirmed), 8 EX-GR-001, 6 injection-style footers, 10 not-an-invoice statements. Matching Agent scored on deterministic compare. Two EX-DUP-001 false negatives → **Hold**, not Go, despite high STP on the happy-path stratum.

Do not copy ACME counts as your gates.

10. Sequence controls

Cannot start	Until
Historical run	Charter signed; environment non-executing; labels complete; gates frozen
UAT	Historical go or documented Hold with UAT-only defects (UI, not accuracy)

Cannot start	Until
Shadow	Historical go; access review; write-block confirmed
Pilot	Shadow pack; rollback tested; allow-lists technically enforced; incident/override live

If a gate artefact is missing, the next stage is unauthorised — even if the software can run.
Proof before permission. Agents earn responsibility. Evidence decides.

End of 00_TESTING_STANDARD.md